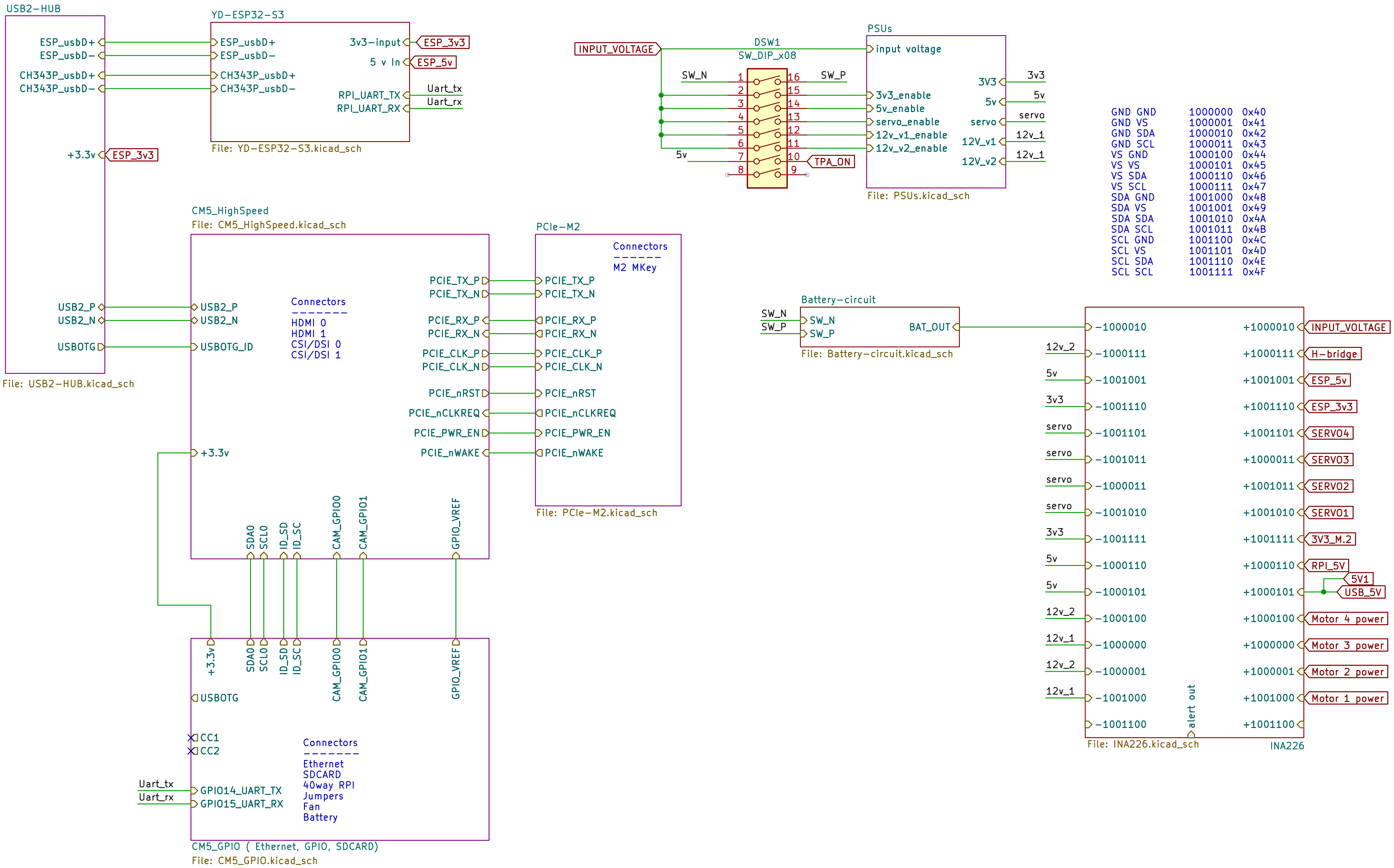
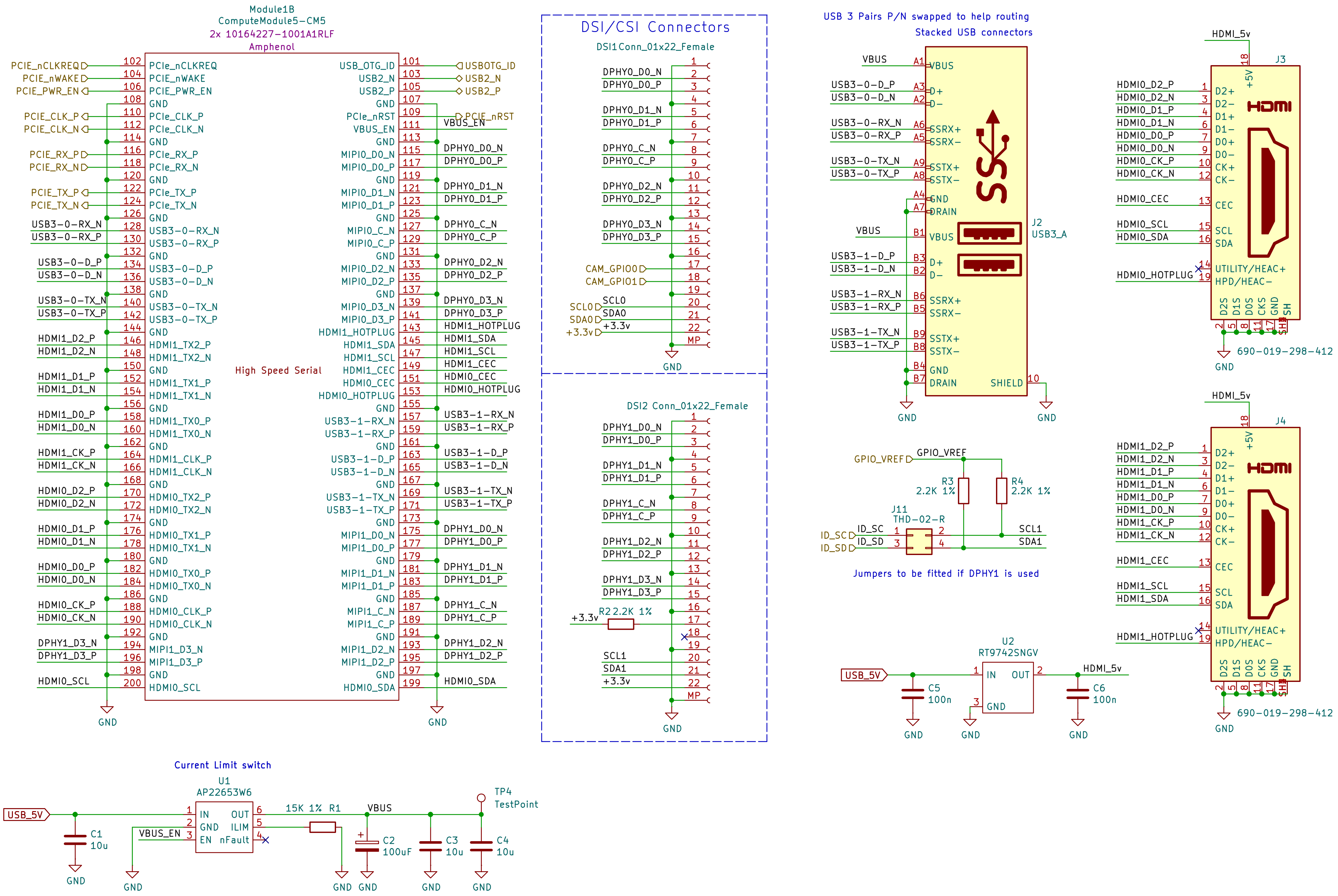
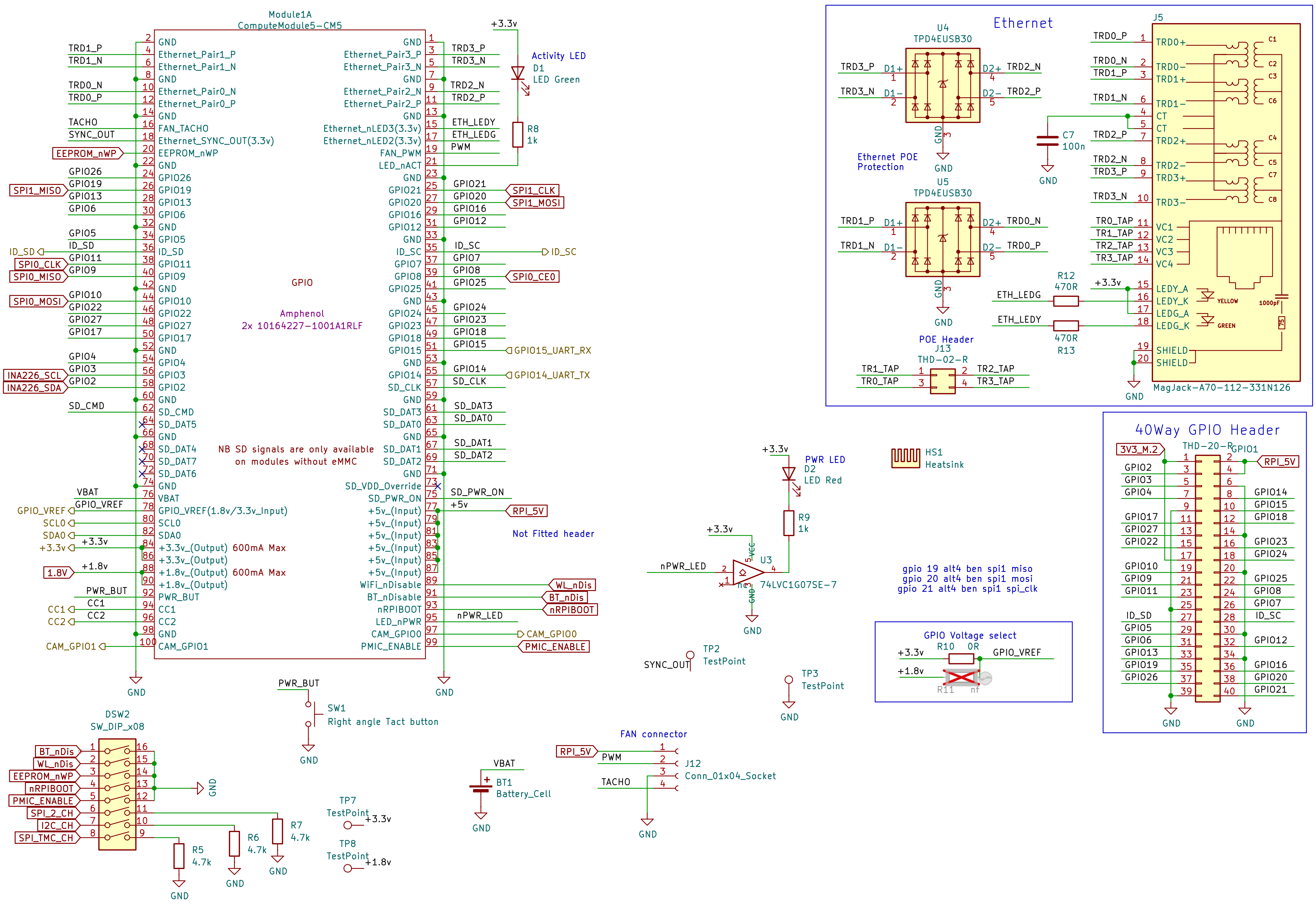
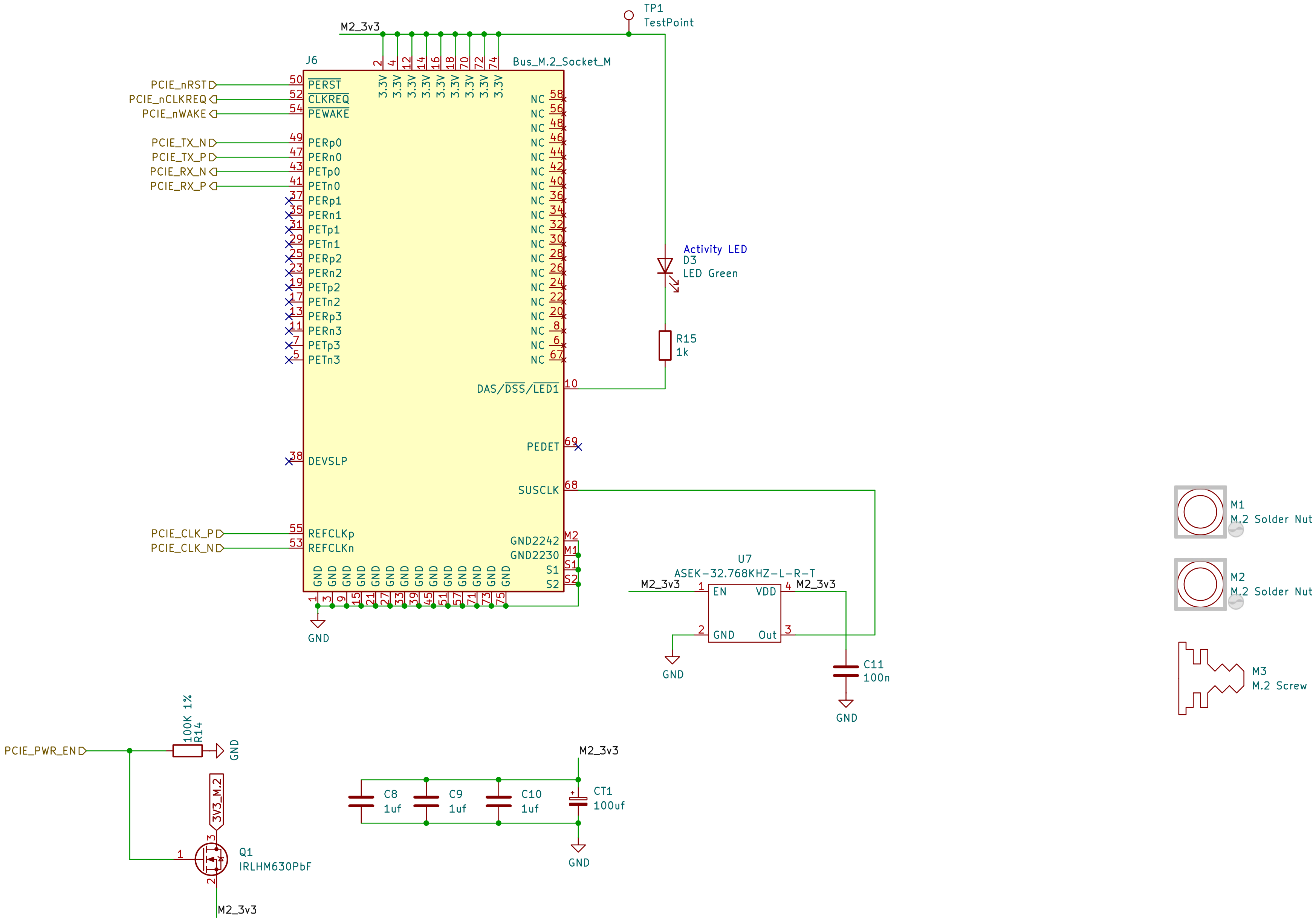


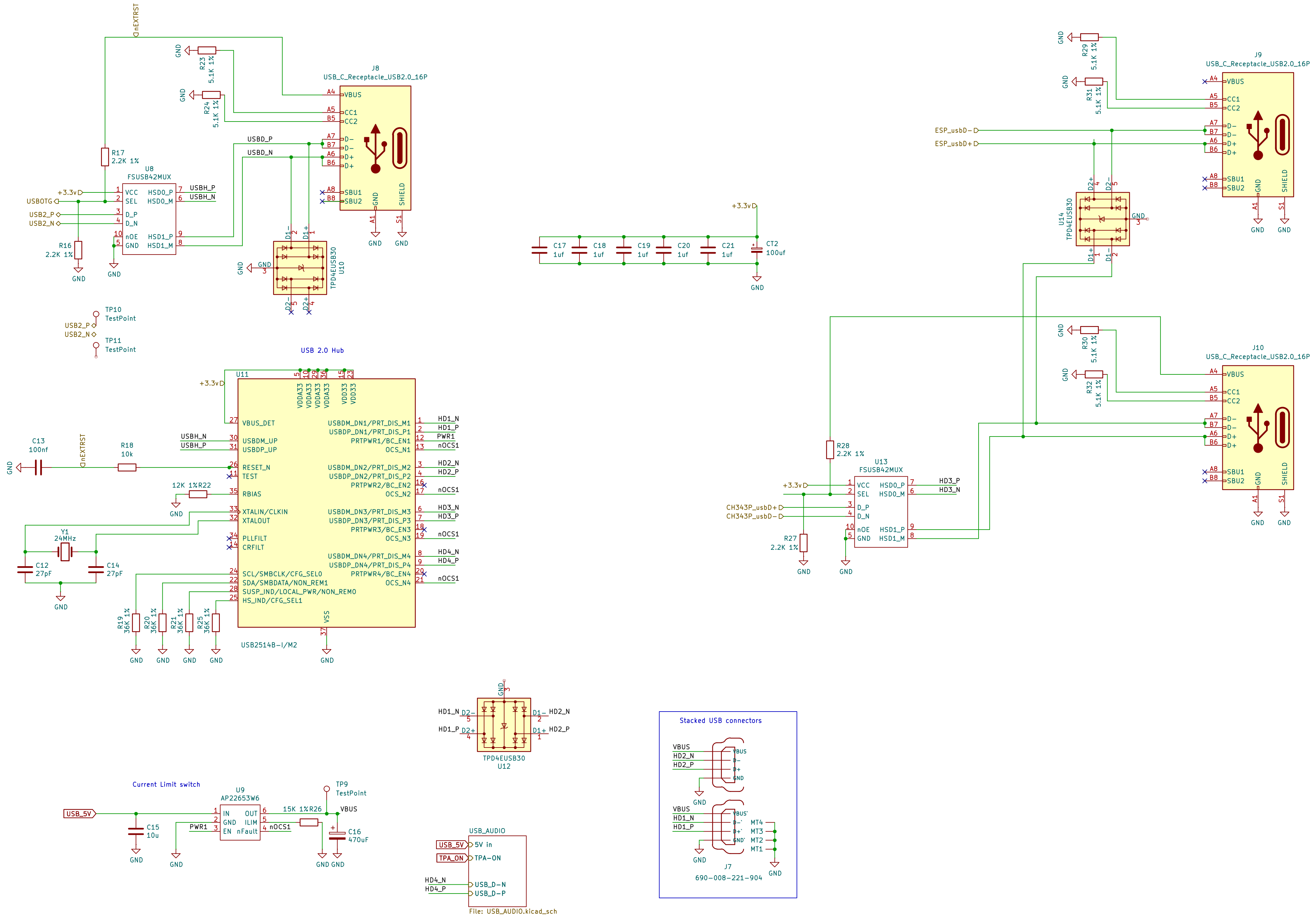
USB2-HUB
ESP_usbD+
ESP_usbD-
CH343P_usbD+
CH343P_usbD-
+3.3v
ESP_3v3
USB2_P
USB2_N
USBOTG
File: USB2-HUB.kicad_sch
YD-ESP32-S3
3v3-input
5 v In
ESP_5v
RPI_UART_TX
RPI_UART_RX
Uart_tx
Uart_rx
File: YD-ESP32-S3.kicad_sch
INPUT_VOLTAGE
DSW1
SW_DIP_x08
SW_N
SW_P
5v
TPA_ON
PSUs
input voltage
3v3_enable
5v_enable
servo_enable
12v_v1_enable
12v_v2_enable
3V3
5v
servo
12V_v1
12V_v2
3v3
12v_1
File: PSUs.kicad_sch
GND GND 1000000 0x40
GND VS 1000001 0x41
GND SDA 1000010 0x42
GND SCL 1000011 0x43
VS GND 1000100 0x44
VS VS 1000101 0x45
VS SDA 1000110 0x46
VS SCL 1000111 0x47
SDA GND 1001000 0x48
SDA VS 1001001 0x49
SDA SDA 1001010 0x4A
SDA SCL 1001011 0x4B
SCL GND 1001100 0x4C
SCL VS 1001101 0x4D
SCL SDA 1001110 0x4E
SCL SCL 1001111 0x4F
CM5_HighSpeed
File: CM5_HighSpeed.kicad_sch
USB2_P
USB2_N
USBOTG_ID
Connectors
HDMI 0
HDMI 1
CSI/DSI 0
CSI/DSI 1
PCIE_TX_P
PCIE_TX_N
PCIE_RX_P
PCIE_RX_N
PCIE_CLK_P
PCIE_CLK_N
PCIE_nRST
PCIE_nCLKREQ
PCIE_PWR_EN
PCIE_nWAKE
+3.3v
SDA0
SCL0
ID_SD
ID_SC
CAM_GPIO0
CAM_GPIO1
GPIO_VREF
PCIe-M2
Connectors
M2 MKey
File: PCIe-M2.kicad_sch
Battery-circuit
SW_N
SW_P
BAT_OUT
File: Battery-circuit.kicad_sch
-1000010
-1000111
-1001001
-1001110
-1001101
-1001011
-1000011
-1001010
-1001111
-1000110
-1000101
-1000100
-1000000
-1000001
-1001000
-1001100
12v_2
5v
3v3
servo
12v_1
+1000010
+1000111
+1001001
+1001110
+1001101
+1000011
+1001011
+1001010
+1001111
+1000110
+1000101
+1000100
+1000000
+1000001
+1001000
+1001100
INPUT_VOLTAGE
H-bridge
ESP_5v
ESP_3v3
SERVO4
SERVO3
SERVO2
SERVO1
3V3_M.2
RPI_5V
5V1
USB_5V
Motor 4 power
Motor 3 power
Motor 2 power
Motor 1 power
alert out
File: INA226.kicad_sch
INA226
USBOTG
CC1
CC2
GPIO14_UART_TX
GPIO15_UART_RX
Connectors
Ethernet
SDCARD
40way RPI
Jumpers
Fan
Battery
CM5_GPIO (Ethernet, GPIO, SDCARD)
File: CM5_GPIO.kicad_sch



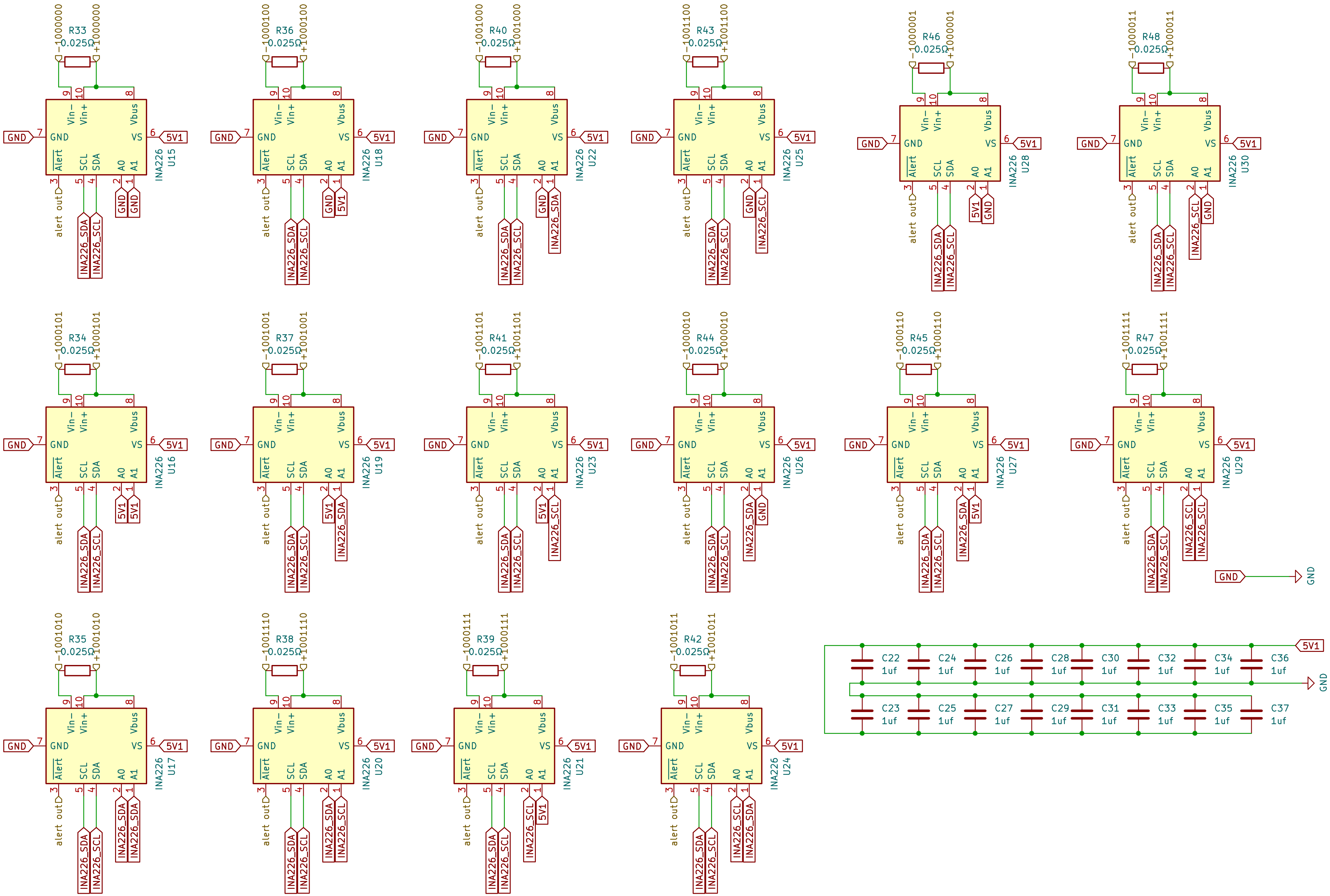
Module1B
ComputeModule5-CM5
2x 10164227-1001A1RLF
Amphenol
High Speed Serial
PCIE_nCLKREQ
PCIE_nWAKE
PCIE_PWR_EN
PCIE_CLK_P
PCIE_CLK_N
PCIE_RX_P
PCIE_RX_N
PCIE_TX_P
PCIE_TX_N
USB3-0-RX_N
USB3-0-RX_P
USB3-0-D_P
USB3-0-D_N
USB3-0-TX_N
USB3-0-TX_P
HDMI1_D2_P
HDMI1_D2_N
HDMI1_D1_P
HDMI1_D1_N
HDMI1_D0_P
HDMI1_D0_N
HDMI1_CK_P
HDMI1_CK_N
HDMI0_D2_P
HDMI0_D2_N
HDMI0_D1_P
HDMI0_D1_N
HDMI0_D0_P
HDMI0_D0_N
HDMI0_CK_P
HDMI0_CK_N
DPHY1_D3_N
DPHY1_D3_P
HDMI0_SCL
USB_OTG_ID
USB2_N
USB2_P
PCIe_nRST
VBUS_EN
USBOTG_ID
PCIE_nRST
DPHY0_D0_N
DPHY0_D0_P
DPHY0_D1_N
DPHY0_D1_P
DPHY0_C_N
DPHY0_C_P
DPHY0_D2_N
DPHY0_D2_P
DPHY0_D3_N
DPHY0_D3_P
HDMI1_HOTPLUG
HDMI1_SDA
HDMI1_SCL
HDMI1_CEC
HDMI0_CEC
HDMI0_HOTPLUG
USB3-1-RX_N
USB3-1-RX_P
USB3-1-D_P
USB3-1-D_N
USB3-1-TX_N
USB3-1-TX_P
DPHY1_D0_N
DPHY1_D0_P
DPHY1_D1_N
DPHY1_D1_P
DPHY1_C_N
DPHY1_C_P
DPHY1_D2_N
DPHY1_D2_P
HDMI0_SDA
GND
DSI/CSI Connectors
DSI1Conn_01x22_Female
CAM_GPIO0
CAM_GPIO1
SCL0
SDA0
+3.3v
MP
DSI2 Conn_01x22_Female
R2 2.2K 1%
SCL1
SDA1
USB 3 Pairs P/N swapped to help routing
Stacked USB connectors
VBUS
J2
USB3_A
SHIELD
GPIO_VREF
R3 2.2K 1%
R4 2.2K 1%
J11
THD-02-R
ID_SC
ID_SD
Jumpers to be fitted if DPHY1 is used
U2
RT9742SNGV
USB_5V
IN OUT GND
HDMI_5v
C5 100n
C6 100n
J3
J4
HDMI0_D2_P
HDMI0_D2_N
HDMI0_D1_P
HDMI0_D1_N
HDMI0_D0_P
HDMI0_D0_N
HDMI0_CK_P
HDMI0_CK_N
HDMI0_CEC
HDMI0_SCL
HDMI0_SDA
HDMI0_HOTPLUG
D2+ D2- D1+ D1- D0+ D0- CK+ CK- CEC SCL SDA
UTILITY/HEAC+
HPD/HEAC-
+5V
D2S D1S D0S CKS GND SH
690-019-298-412
HDMI1_D2_P
HDMI1_D2_N
HDMI1_D1_P
HDMI1_D1_N
HDMI1_D0_P
HDMI1_D0_N
HDMI1_CK_P
HDMI1_CK_N
HDMI1_CEC
HDMI1_SCL
HDMI1_SDA
HDMI1_HOTPLUG
Current Limit switch
U1
AP22653W6
IN GND EN OUT ILIM nFault
15K 1% R1
C1 10u
C2 100uF
C3 10u
C4 10u
TP4
TestPoint

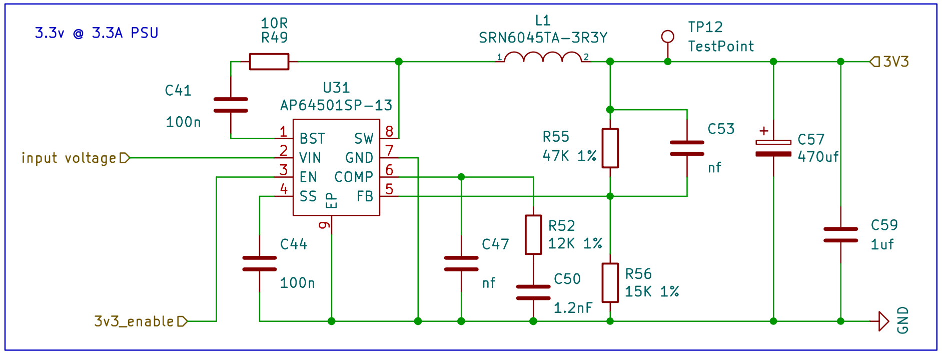

Module1A
ComputeModule5-CM5
GPIO
Amphenol
2x 10164227-1001A1RLF
NB SD signals are only available
on modules without eMMC
GND
Ethernet_Pair1_P
Ethernet_Pair1_N
Ethernet_Pair0_N
Ethernet_Pair0_P
FAN_TACHO
Ethernet_SYNC_OUT(3.3v)
EEPROM_nWP
GPIO26
GPIO19
GPIO13
GPIO6
GPIO5
ID_SD
GPIO11
GPIO9
GPIO10
GPIO22
GPIO27
GPIO17
GPIO4
GPIO3
GPIO2
SD_CMD
SD_DAT5
SD_DAT4
SD_DAT7
SD_DAT6
VBAT
GPIO_VREF(1.8v/3.3v_Input)
SCL0
SDA0
+3.3v_(Output) 600mA Max
+3.3v_(Output)
+1.8v_(Output) 600mA Max
+1.8v_(Output)
PWR_BUT
CC1
CC2
CAM_GPIO1
Ethernet_Pair3_P
Ethernet_Pair3_N
Ethernet_Pair2_N
Ethernet_Pair2_P
Ethernet_nLED3(3.3v)
Ethernet_nLED2(3.3v)
FAN_PWM
LED_nACT
GPIO21
GPIO20
GPIO16
GPIO12
ID_SC
GPIO7
GPIO8
GPIO25
GPIO24
GPIO23
GPIO18
GPIO15
GPIO14
SD_CLK
SD_DAT3
SD_DAT0
SD_DAT1
SD_DAT2
SD_VDD_Override
SD_PWR_ON
+5v_(Input)
WiFi_nDisable
BT_nDisable
nRPIBOOT
LED_nPWR
CAM_GPIO0
PMIC_ENABLE
TRD1_P
TRD1_N
TRD0_N
TRD0_P
TACHO
SYNC_OUT
EEPROM_nWP
SPI1_MISO
SPI0_CLK
SPI0_MISO
SPI0_MOSI
INA226_SCL
INA226_SDA
GPIO_VREF
+3.3v
+1.8v
1.8V
TRD3_P
TRD3_N
TRD2_N
TRD2_P
ETH_LEDY
ETH_LEDG
PWM
SPI1_CLK
SPI1_MOSI
SPI0_CE0
GPIO15_UART_RX
GPIO14_UART_TX
+5v
RPI_5V
Not Fitted header
WL_nDis
BT_nDis
nRPIBOOT
nPWR_LED
CAM_GPIO0
PMIC_ENABLE
Activity LED
D1
LED Green
R8
1k
Ethernet
U4
TPD4EUSB30
U5
TPD4EUSB30
Ethernet POE
Protection
C7
100n
J5
MagJack-A70-112-331N126
TRD0+
TRD0-
TRD1+
TRD1-
CT
TRD2+
TRD2-
TRD3+
TRD3-
VC1
VC2
VC3
VC4
LEDY_A
LEDY_K
LEDG_A
LEDG_K
SHIELD
YELLOW
GREEN
1000pF
R12
470R
R13
470R
POE Header
J13
THD-02-R
TR0_TAP
TR1_TAP
TR2_TAP
TR3_TAP
40Way GPIO Header
THD-20-R
GPIO1
3V3_M.2
HS1
Heatsink
PWR LED
D2
LED Red
R9
1k
U3
74LVC1G07SE-7
gpio 19 alt4 ben spi1 miso
gpio 20 alt4 ben spi1 mosi
gpio 21 alt4 ben spi1 spi_clk
GPIO Voltage select
R10 0R
R11 nf
TP2
TestPoint
TP3
TestPoint
DSW2
SW_DIP_x08
BT_nDis
WL_nDis
EEPROM_nWP
nRPIBOOT
PMIC_ENABLE
SPI_2_CH
I2C_CH
SPI_TMC_CH
R5
4.7k
R6
4.7k
R7
4.7k
SW1
Right angle Tact button
TP7
TestPoint
TP8
TestPoint
BT1
Battery_Cell
FAN connector
J12
Conn_01x04_Socket

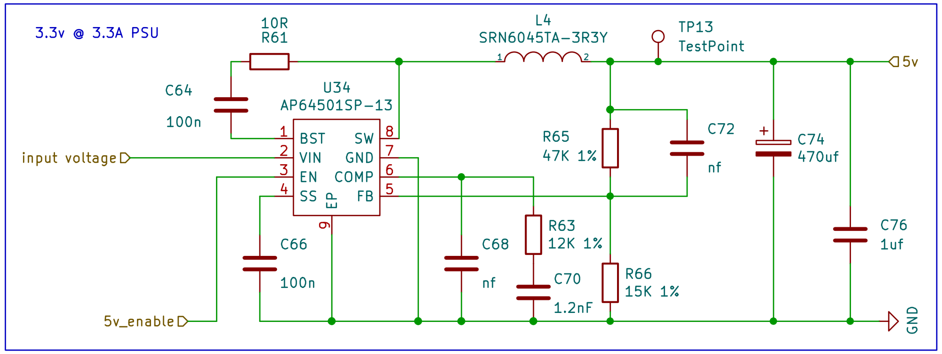

M2_3v3
TP1
TestPoint
J6
Bus_M.2_Socket_M
PCIE_nRST
PCIE_nCLKREQ
PCIE_nWAKE
PCIE_TX_N
PCIE_TX_P
PCIE_RX_N
PCIE_RX_P
PERST
CLKREQ
PEWAKE
PERp0
PERn0
PETp0
PETn0
PERp1
PERn1
PETp1
PETn1
PERp2
PERn2
PETp2
PETn2
PERp3
PERn3
PETp3
PETn3
DEVSLP
REFCLKp
REFCLKn
PCIE_CLK_P
PCIE_CLK_N
3.3V
NC
DAS/DSS/LED1
PEDET
SUSCLK
GND2242
GND2230
S1
S2
GND
Activity LED
D3
LED Green
R15
1k
U7
ASEK-32.768KHZ-L-R-T
EN
VDD
Out
C11
100n
M1
M.2 Solder Nut
M2
M.2 Solder Nut
M3
M.2 Screw
PCIE_PWR_EN
100K 1%
R14
3V3_M.2
Q1
IRLHM630PbF
C8
1uf
C9
1uf
C10
1uf
CT1
100uf

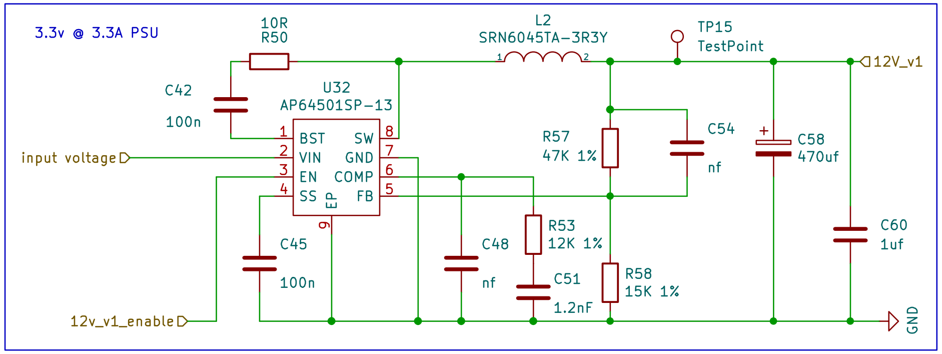


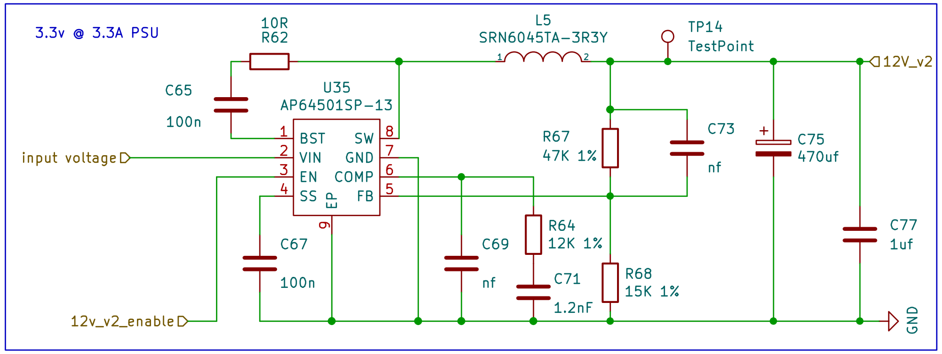
nEXTRST
GND
R23
5.1K 1%
R24
5.1K 1%
J8
USB_C_Receptacle_USB2.0_16P
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
SHIELD
USBD_P
USBD_N
R17
2.2K 1%
U8
FSUSB42MUX
+3.3v
USBOTG
USB2_P
USB2_N
VCC
SEL
D_P
D_N
nOE
GND
HSD0_P
HSD0_M
HSD1_P
HSD1_M
USBH_P
USBH_N
R16
2.2K 1%
TPD4EUSB30
U10
TP10
TestPoint
TP11
TestPoint
USB 2.0 Hub
U11
VBUS_DET
USBDM_UP
USBDP_UP
RESET_N
TEST
RBIAS
XTALIN/CLKIN
XTALOUT
PLLFILT
CRFILT
SCL/SMBCLK/CFG_SEL0
SDA/SMBDATA/NON_REM1
SUSP_IND/LOCAL_PWR/NON_REM0
HS_IND/CFG_SEL1
VDDA33
VDD33
VSS
USBDM_DN1/PRT_DIS_M1
USBDP_DN1/PRT_DIS_P1
PRTPWR1/BC_EN1
OCS_N1
USBDM_DN2/PRT_DIS_M2
USBDP_DN2/PRT_DIS_P2
PRTPWR2/BC_EN2
OCS_N2
USBDM_DN3/PRT_DIS_M3
USBDP_DN3/PRT_DIS_P3
PRTPWR3/BC_EN3
OCS_N3
USBDM_DN4/PRT_DIS_M4
USBDP_DN4/PRT_DIS_P4
PRTPWR4/BC_EN4
OCS_N4
USB2514B-I/M2
HD1_N
HD1_P
PWR1
nOCS1
HD2_N
HD2_P
HD3_N
HD3_P
HD4_N
HD4_P
C13
100nf
R18
10k
12K 1%R22
Y1
24MHz
C12
27pF
C14
27pF
R19
36K 1%
R20
36K 1%
R21
36K 1%
R25
36K 1%
C17
1uf
C18
1uf
C19
1uf
C20
1uf
C21
1uf
CT2
100uf
R29
5.1K 1%
R31
5.1K 1%
J9
USB_C_Receptacle_USB2.0_16P
ESP_usbD-
ESP_usbD+
U14
TPD4EUSB30
R30
5.1K 1%
R32
5.1K 1%
J10
USB_C_Receptacle_USB2.0_16P
R28
2.2K 1%
U13
FSUSB42MUX
CH343P_usbD+
CH343P_usbD-
R27
2.2K 1%
TPD4EUSB30
U12
Stacked USB connectors
VBUS
J7
690-008-221-904
MT1
MT2
MT3
MT4
Current Limit switch
U9
AP22653W6
USB_5V
IN
GND
EN
OUT
ILIM
nFault
C15
10u
15K 1%R26
TP9
TestPoint
C16
470uF
USB_AUDIO
5V in
TPA-ON
TPA_ON
USB_D-N
USB_D-P
File: USB_AUDIO.kicad_sch

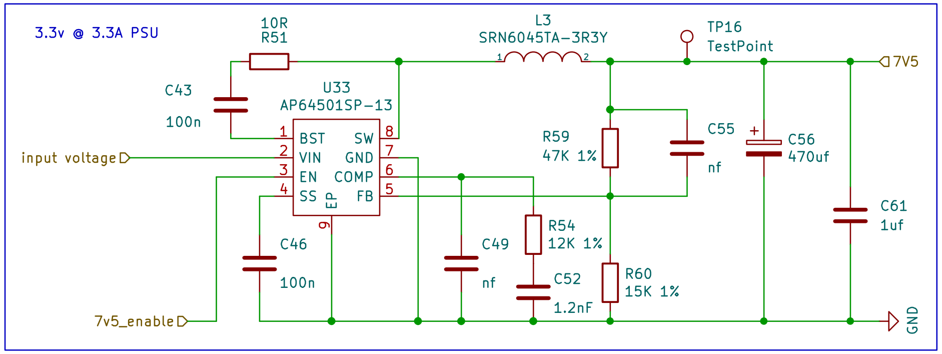

R33 0.025Ω -1000000 +1000000
U15 INA226
R36 0.025Ω -1000100 +1000100
U18 INA226
R40 0.025Ω -1001000 +1001000
U22 INA226
R43 0.025Ω -1001100 +1001100
U25 INA226
R46 0.025Ω -1000001 +1000001
U28 INA226
R48 0.025Ω -1000011 +1000011
U30 INA226
R34 0.025Ω -1000101 +1000101
U16 INA226
R37 0.025Ω -1001001 +1001001
U19 INA226
R41 0.025Ω -1001101 +1001101
U23 INA226
R44 0.025Ω -1000010 +1000010
U26 INA226
R45 0.025Ω -1000110 +1000110
U27 INA226
R47 0.025Ω -1001111 +1001111
U29 INA226
R35 0.025Ω -1001010 +1001010
U17 INA226
R38 0.025Ω -1001110 +1001110
U20 INA226
R39 0.025Ω -1000111 +1000111
U21 INA226
R42 0.025Ω -1001011 +1001011
U24 INA226
Vin- Vin+ Vbus GND VS Alert SCL SDA A0 A1
GND 5V1 alert out INA226_SDA INA226_SCL
C22 1uf C24 1uf C26 1uf C28 1uf C30 1uf C32 1uf C34 1uf C36 1uf
C23 1uf C25 1uf C27 1uf C29 1uf C31 1uf C33 1uf C35 1uf C37 1uf

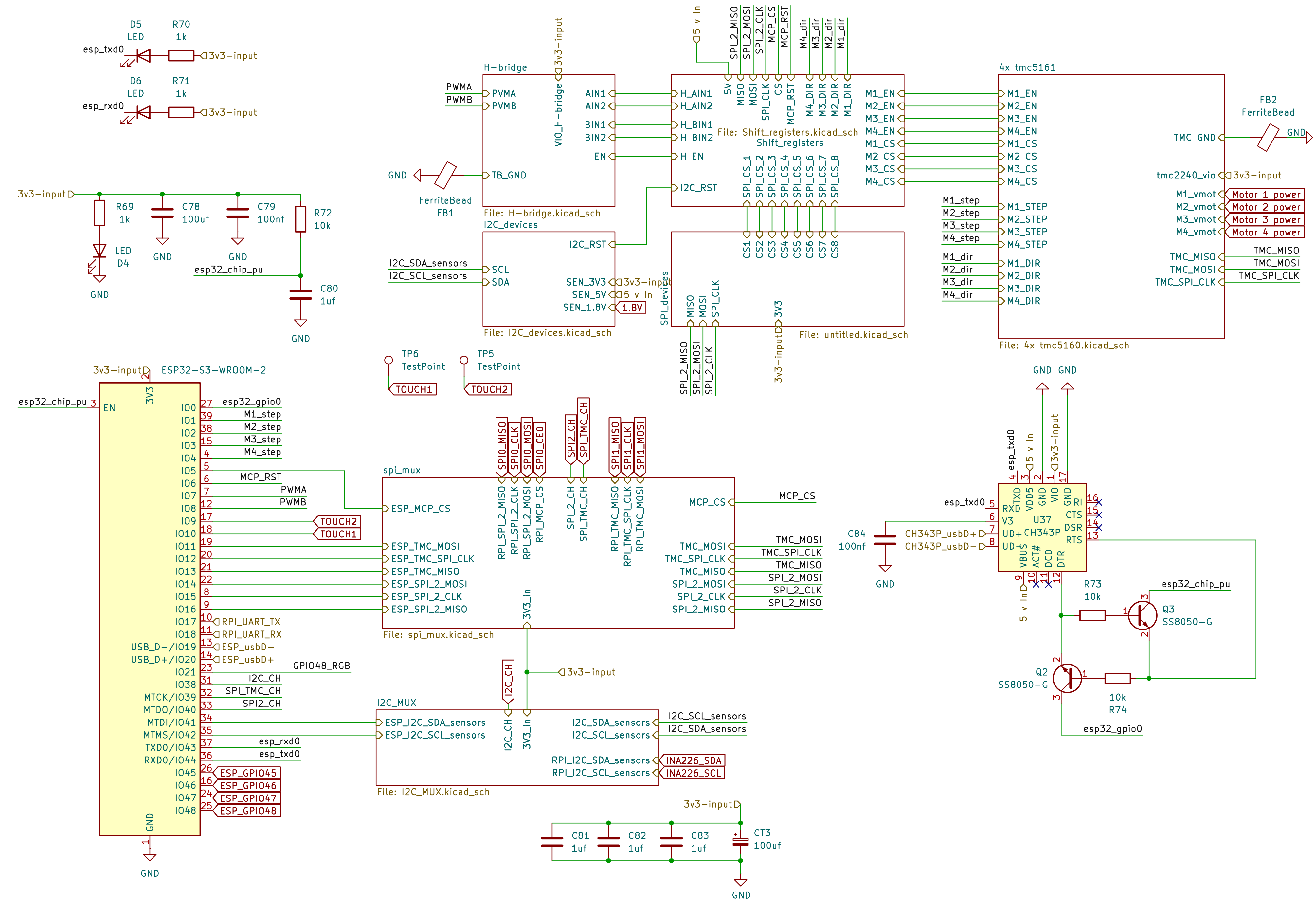

3.3v @ 3.3A PSU
10R
R49
C41
100n
U31
AP64501SP-13
1 BST
2 VIN
3 EN
4 SS
SW 8
GND 7
COMP 6
FB 5
EP 9
input voltage
3v3_enable
C44
100n
L1
SRN6045TA-3R3Y
TP12
TestPoint
R55
47K 1%
C53
nf
C57
470uf
3V3
C47
nf
R52
12K 1%
C50
1.2nF
R56
15K 1%
C59
1uf
GND


3.3v @ 3.3A PSU
10R
R61
C64
100n
U34
AP64501SP-13
1 BST
2 VIN
3 EN
4 SS
SW 8
GND 7
COMP 6
FB 5
EP 9
input voltage
5v_enable
C66
100n
L4
SRN6045TA-3R3Y
TP13
TestPoint
R65
47K 1%
C72
nf
C74
470uf
5v
C68
nf
R63
12K 1%
C70
1.2nF
R66
15K 1%
C76
1uf
GND

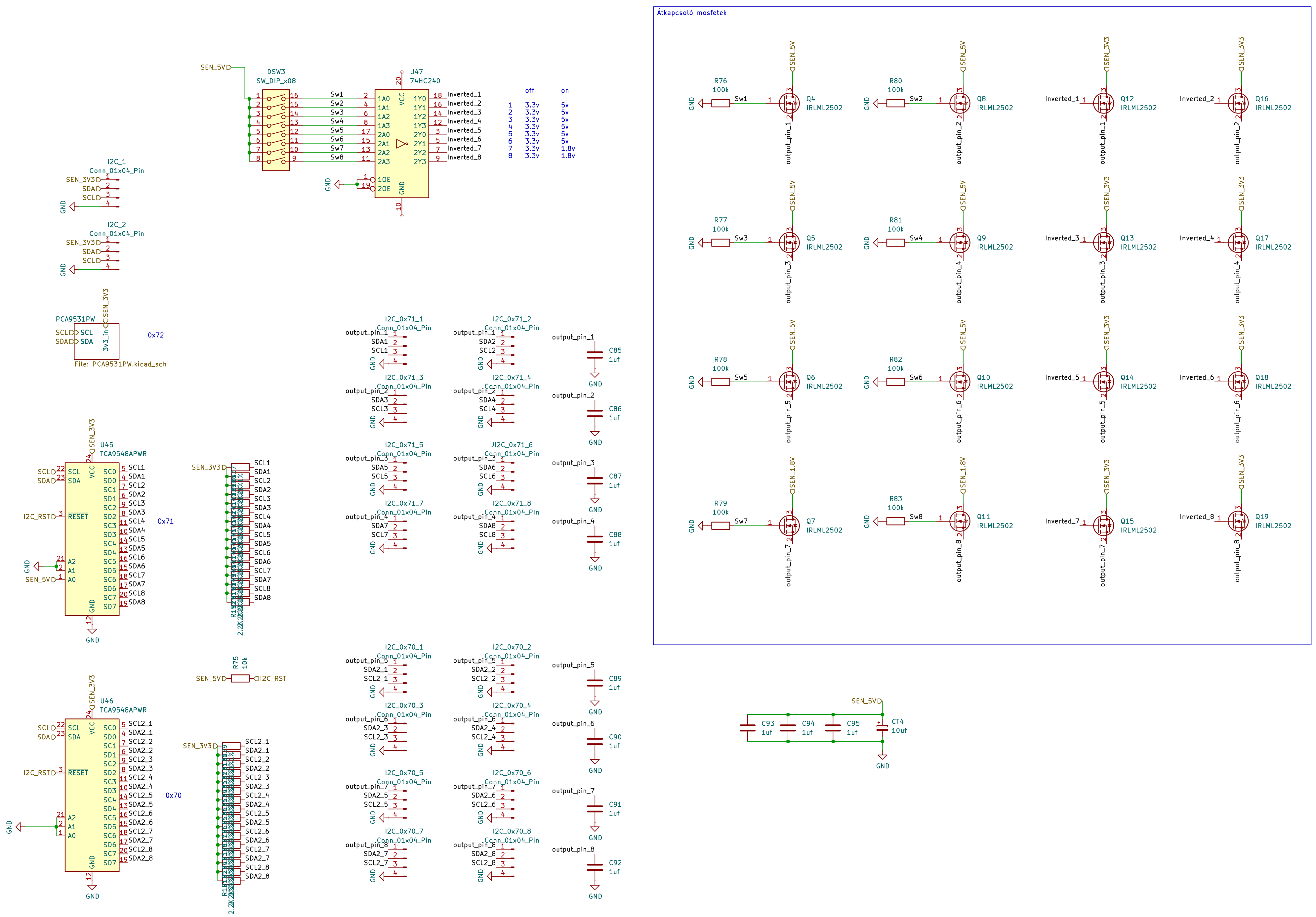

3.3v @ 3.3A PSU
10R
R50
C42
100n
U32
AP64501SP-13
1 BST
2 VIN
3 EN
4 SS
SW 8
GND 7
COMP 6
FB 5
EP 9
input voltage
12v_v1_enable
C45
100n
L2
SRN6045TA-3R3Y
TP15
TestPoint
R57
47K 1%
C54
nf
C58
470uf
12V_v1
C48
nf
R53
12K 1%
C51
1.2nF
R58
15K 1%
C60
1uf
GND


3.3v @ 3.3A PSU
10R
R62
C65
100n
U35
AP64501SP-13
1 BST
2 VIN
3 EN
4 SS
SW 8
GND 7
COMP 6
FB 5
EP 9
input voltage
12v_v2_enable
C67
100n
L5
SRN6045TA-3R3Y
TP14
TestPoint
R67
47K 1%
C73
nf
C75
470uf
12V_v2
C69
nf
R64
12K 1%
C71
1.2nF
R68
15K 1%
C77
1uf
GND


3.3v @ 3.3A PSU
10R
R51
C43
100n
U33
AP64501SP-13
1 BST
2 VIN
3 EN
4 SS
SW 8
GND 7
COMP 6
FB 5
EP 9
input voltage
7v5_enable
C46
100n
L3
SRN6045TA-3R3Y
TP16
TestPoint
R59
47K 1%
C55
nf
C56
470uf
7V5
C49
nf
R54
12K 1%
C52
1.2nF
R60
15K 1%
C61
1uf
GND

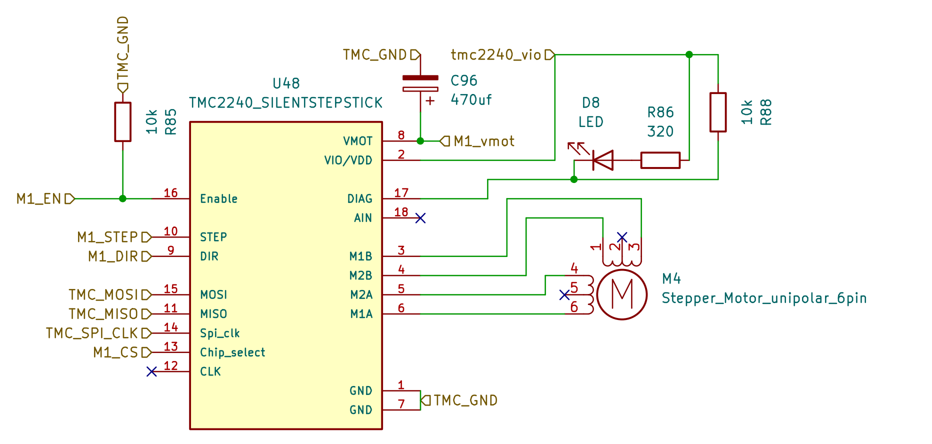


D5
LED
R70
1k
esp_txd0
3v3-input
D6
LED
R71
1k
esp_rxd0
3v3-input
3v3-input
R69
1k
C78
100uf
C79
100nf
R72
10k
LED
D4
GND
GND
GND
esp32_chip_pu
C80
1uf
GND
H-bridge
PWMA
PWMB
PVMA
PVMB
VIO_H-bridge
3v3-input
AIN1
AIN2
BIN1
BIN2
EN
GND
TB_GND
FerriteBead
FB1
File: H-bridge.kicad_sch
I2C_devices
I2C_RST
I2C_SDA_sensors
I2C_SCL_sensors
SCL
SDA
SEN_3V3
SEN_5V
SEN_1.8V
3v3-input
5 v In
1.8V
File: I2C_devices.kicad_sch
5 v In
SPI_2_MISO
SPI_2_MOSI
SPI_2_CLK
MCP_CS
MCP_RST
M4_dir
M3_dir
M2_dir
M1_dir
5V
MISO
MOSI
SPI_CLK
CS
MCP_RST
M4_DIR
M3_DIR
M2_DIR
M1_DIR
H_AIN1
H_AIN2
H_BIN1
H_BIN2
H_EN
I2C_RST
File: Shift_registers.kicad_sch
Shift_registers
SPI_CS_1
SPI_CS_2
SPI_CS_3
SPI_CS_4
SPI_CS_5
SPI_CS_6
SPI_CS_7
SPI_CS_8
M1_EN
M2_EN
M3_EN
M4_EN
M1_CS
M2_CS
M3_CS
M4_CS
CS1
CS2
CS3
CS4
CS5
CS6
CS7
CS8
SPI_devices
MISO
MOSI
SPI_CLK
3V3
File: untitled.kicad_sch
SPI_2_MISO
SPI_2_MOSI
SPI_2_CLK
3v3-input
4x tmc5161
M1_EN
M2_EN
M3_EN
M4_EN
M1_CS
M2_CS
M3_CS
M4_CS
M1_step
M2_step
M3_step
M4_step
M1_STEP
M2_STEP
M3_STEP
M4_STEP
M1_dir
M2_dir
M3_dir
M4_dir
M1_DIR
M2_DIR
M3_DIR
M4_DIR
TMC_GND
FB2
FerriteBead
GND
tmc2240_vio
3v3-input
M1_vmot
M2_vmot
M3_vmot
M4_vmot
Motor 1 power
Motor 2 power
Motor 3 power
Motor 4 power
TMC_MISO
TMC_MOSI
TMC_SPI_CLK
TMC_MISO
TMC_MOSI
TMC_SPI_CLK
File: 4x tmc5160.kicad_sch
TP6
TestPoint
TOUCH1
TP5
TestPoint
TOUCH2
3v3-input
ESP32-S3-WROOM-2
esp32_chip_pu
EN
3V3
IO0
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8
IO9
IO10
IO11
IO12
IO13
IO14
IO15
IO16
IO17
IO18
USB_D-/IO19
USB_D+/IO20
IO21
IO38
MTCK/IO39
MTDO/IO40
MTDI/IO41
MTMS/IO42
TXD0/IO43
RXD0/IO44
IO45
IO46
IO47
IO48
GND
GND
esp32_gpio0
M1_step
M2_step
M3_step
M4_step
MCP_RST
PWMA
PWMB
TOUCH2
TOUCH1
RPI_UART_TX
RPI_UART_RX
ESP_usbD-
ESP_usbD+
GPIO48_RGB
I2C_CH
SPI_TMC_CH
SPI2_CH
esp_rxd0
esp_txd0
ESP_GPIO45
ESP_GPIO46
ESP_GPIO47
ESP_GPIO48
SPI0_MISO
SPI0_CLK
SPI0_MOSI
SPI0_CE0
SPI2_CH
SPI_TMC_CH
SPI1_MISO
SPI1_CLK
SPI1_MOSI
spi_mux
ESP_MCP_CS
ESP_TMC_MOSI
ESP_TMC_SPI_CLK
ESP_TMC_MISO
ESP_SPI_2_MOSI
ESP_SPI_2_CLK
ESP_SPI_2_MISO
RPI_SPI_2_MISO
RPI_SPI_2_CLK
RPI_SPI_2_MOSI
RPI_MCP_CS
SPI_2_CH
SPI_TMC_CH
RPI_TMC_MISO
RPI_TMC_SPI_CLK
RPI_TMC_MOSI
3V3_in
MCP_CS
TMC_MOSI
TMC_SPI_CLK
TMC_MISO
SPI_2_MOSI
SPI_2_CLK
SPI_2_MISO
MCP_CS
TMC_MOSI
TMC_SPI_CLK
TMC_MISO
SPI_2_MOSI
SPI_2_CLK
SPI_2_MISO
File: spi_mux.kicad_sch
3v3-input
I2C_CH
I2C_MUX
ESP_I2C_SDA_sensors
ESP_I2C_SCL_sensors
I2C_CH
3V3_in
I2C_SDA_sensors
I2C_SCL_sensors
I2C_SCL_sensors
I2C_SDA_sensors
RPI_I2C_SDA_sensors
RPI_I2C_SCL_sensors
INA226_SDA
INA226_SCL
File: I2C_MUX.kicad_sch
3v3-input
C81
1uf
C82
1uf
C83
1uf
CT3
100uf
GND
GND
GND
esp_txd0
5 v In
3v3-input
esp_txd0
RXD
TXD
VDD5
GND
VIO
GND
V3
U37
CH343P
UD+
UD-
RI
CTS
DSR
RTS
VBUS
ACT#
DCD
DTR
C84
100nf
CH343P_usbD+
CH343P_usbD-
GND
5 v In
R73
10k
esp32_chip_pu
Q3
SS8050-G
Q2
SS8050-G
10k
R74
esp32_gpio0

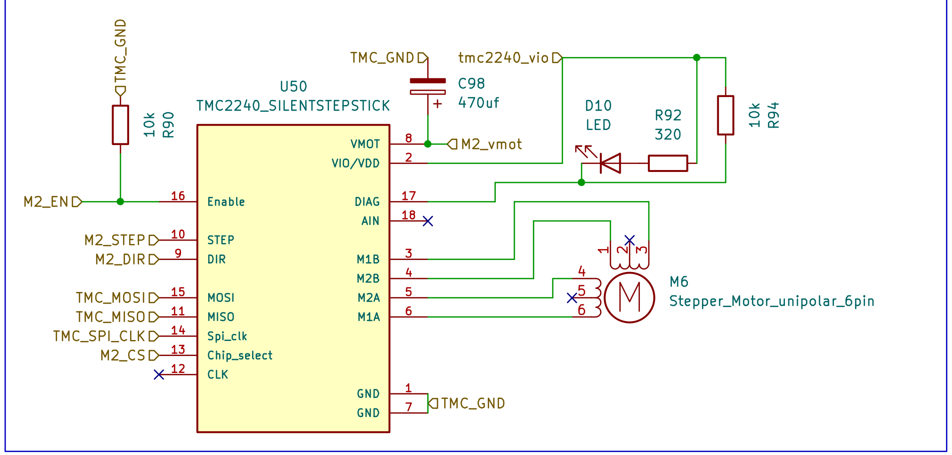


Átkapcsoló mosfetek
SEN_5V
DSW3
SW_DIP_x08
U47
74HC240
Sw1
Sw2
Sw3
Sw4
Sw5
Sw6
Sw7
Sw8
Inverted_1
Inverted_2
Inverted_3
Inverted_4
Inverted_5
Inverted_6
Inverted_7
Inverted_8
off on
1 3.3v 5v
2 3.3v 5v
3 3.3v 5v
4 3.3v 5v
5 3.3v 5v
6 3.3v 5v
7 3.3v 1.8v
8 3.3v 1.8v
I2C_1
Conn_01x04_Pin
I2C_2
Conn_01x04_Pin
SEN_3V3
SDA
SCL
GND
PCA9531PW
File: PCA9531PW.kicad_sch
0x72
U45
TCA9548APWR
0x71
U46
TCA9548APWR
0x70
I2C_RST
R75
10k
R76 100k
R77 100k
R78 100k
R79 100k
R80 100k
R81 100k
R82 100k
R83 100k
Q4 IRLML2502
Q5 IRLML2502
Q6 IRLML2502
Q7 IRLML2502
Q8 IRLML2502
Q9 IRLML2502
Q10 IRLML2502
Q11 IRLML2502
Q12 IRLML2502
Q13 IRLML2502
Q14 IRLML2502
Q15 IRLML2502
Q16 IRLML2502
Q17 IRLML2502
Q18 IRLML2502
Q19 IRLML2502
SEN_1.8V
output_pin_1
output_pin_2
output_pin_3
output_pin_4
output_pin_5
output_pin_6
output_pin_7
output_pin_8
I2C_0x71_1
I2C_0x71_2
I2C_0x71_3
I2C_0x71_4
I2C_0x71_5
JI2C_0x71_6
I2C_0x71_7
I2C_0x71_8
I2C_0x70_1
I2C_0x70_2
I2C_0x70_3
I2C_0x70_4
I2C_0x70_5
I2C_0x70_6
I2C_0x70_7
I2C_0x70_8
C85 1uf
C86 1uf
C87 1uf
C88 1uf
C89 1uf
C90 1uf
C91 1uf
C92 1uf
C93 1uf
C94 1uf
C95 1uf
CT4 10uf

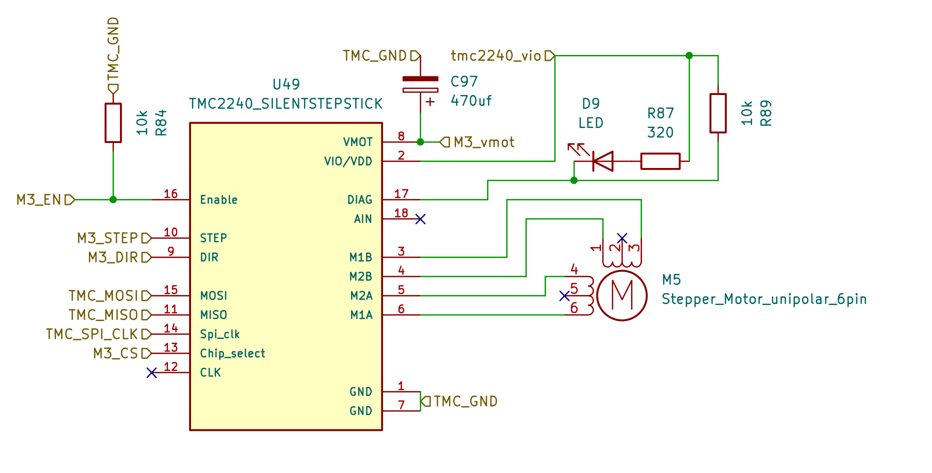
M1_control with TMC2240/TMC5160



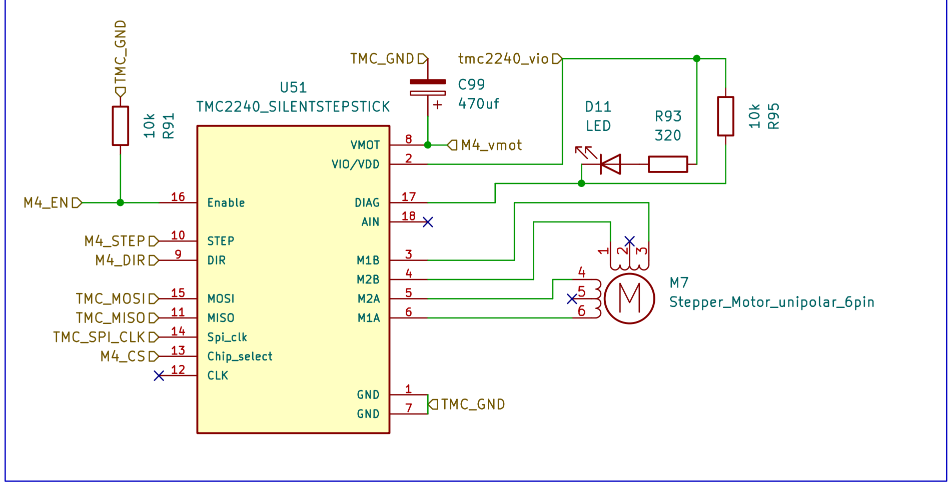
M2_control with TMC2240/TMC5160

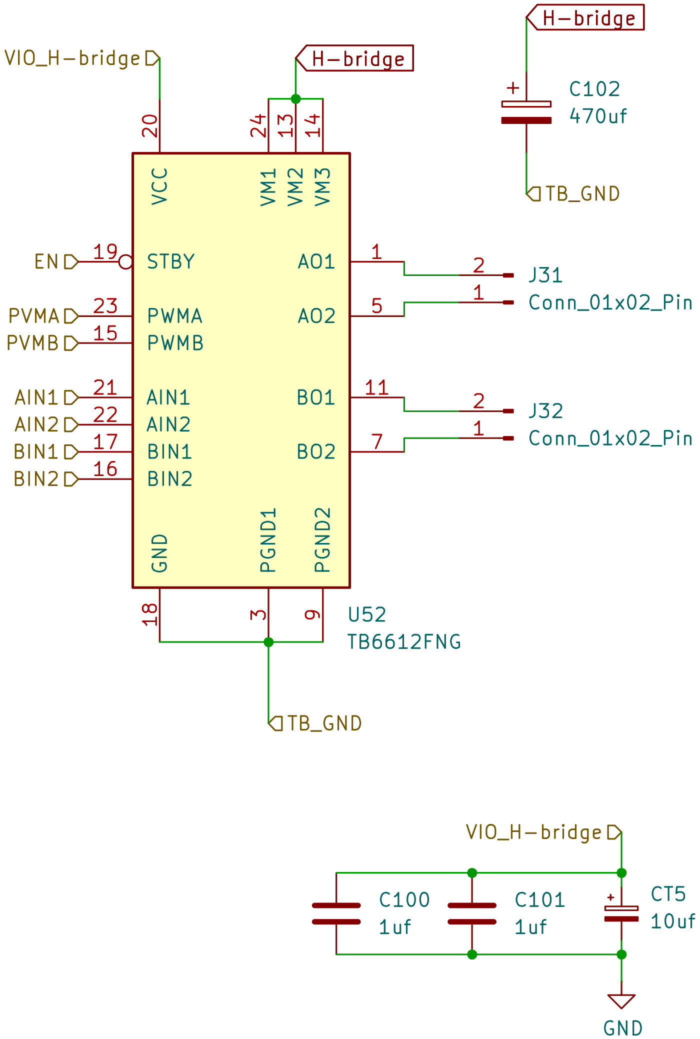


M3_control with TMC2240/TMC5160



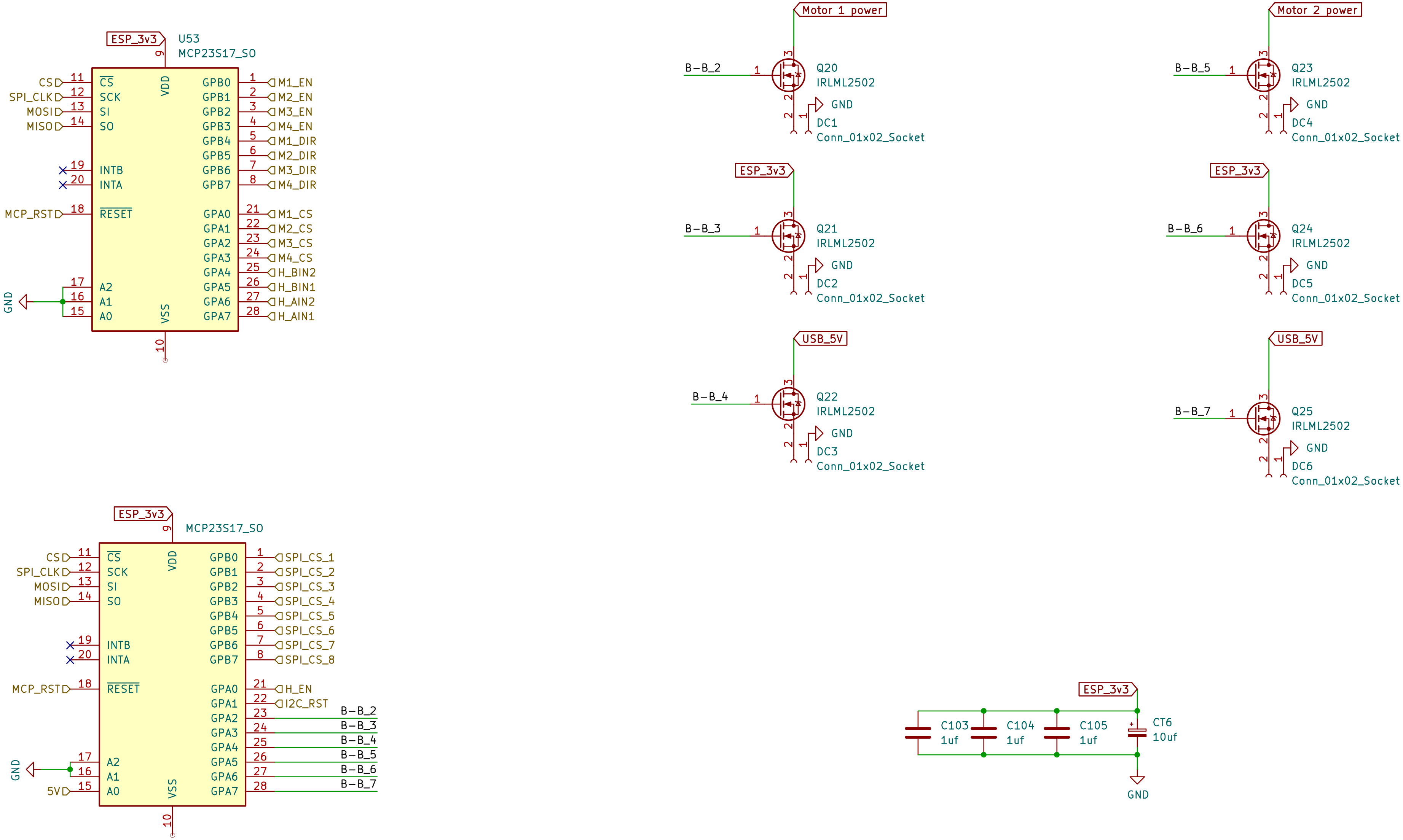
M4_control with TMC2240/TMC5160

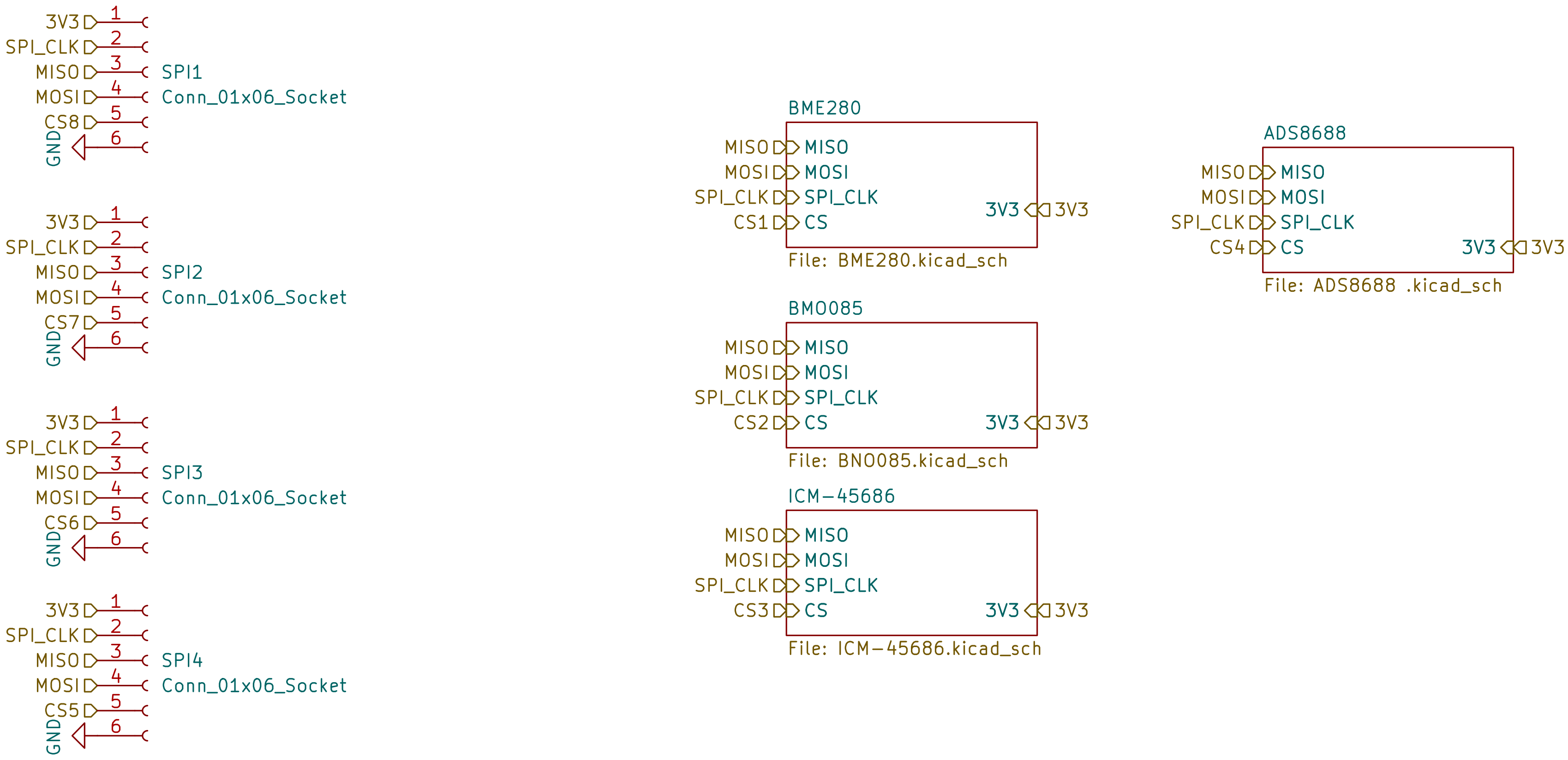


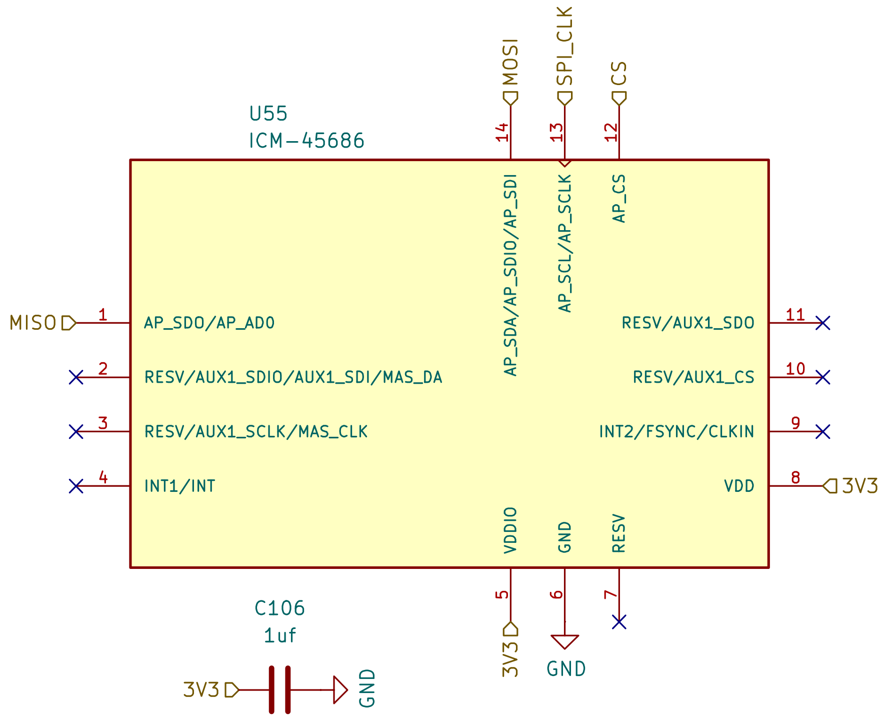



```
//motor A connected between A01 and A02
//motor B connected between B01 and B02

int STBY = 10; //standby

//Motor A
int PWMA = 3; //Speed control
int AIN1 = 9; //Direction
int AIN2 = 8; //Direction

//Motor B
int PWMB = 5; //Speed control
int BIN1 = 11; //Direction
int BIN2 = 12; //Direction

void setup(){
  pinMode(STBY, OUTPUT);

  pinMode(PWMA, OUTPUT);
  pinMode(AIN1, OUTPUT);
  pinMode(AIN2, OUTPUT);

  pinMode(PWMB, OUTPUT);
  pinMode(BIN1, OUTPUT);
  pinMode(BIN2, OUTPUT);
}

void loop(){
  move(1, 255, 1); //motor 1, full speed, left
  move(2, 255, 1); //motor 2, full speed, left

  delay(1000); //go for 1 second
  stop(); //stop
  delay(250); //hold for 250ms until move again

  move(1, 128, 0); //motor 1, half speed, right
  move(2, 128, 0); //motor 2, half speed, right

  delay(1000);
  stop();
  delay(250);
}
```

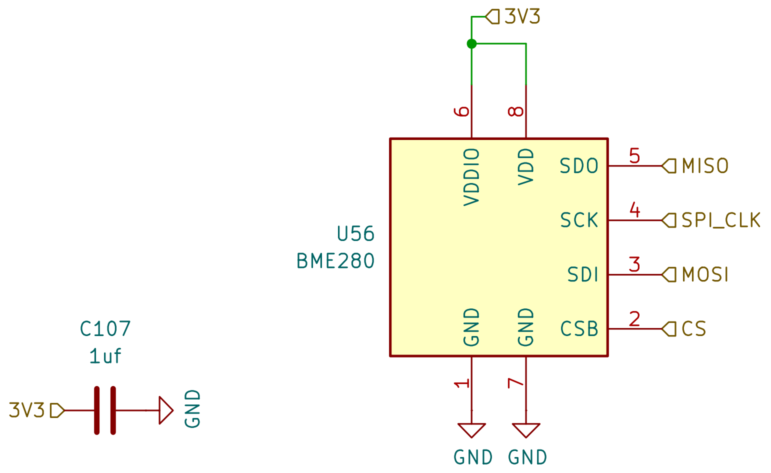
```
void move(int motor, int speed, int direction){
//Move specific motor at speed and direction
//motor: 0 for B 1 for A
//speed: 0 is off, and 255 is full speed
//direction: 0 clockwise, 1 counter-clockwise

  digitalWrite(STBY, HIGH); //disable standby

  boolean inPin1 = LOW;
  boolean inPin2 = HIGH;

  if(direction == 1){
    inPin1 = HIGH;
    inPin2 = LOW;
  }

  if(motor == 1){
    digitalWrite(AIN1, inPin1);
    digitalWrite(AIN2, inPin2);
    analogWrite(PWMA, speed);
  }else{
    digitalWrite(BIN1, inPin1);
    digitalWrite(BIN2, inPin2);
    analogWrite(PWMB, speed);
  }
}

void stop(){
//enable standby
  digitalWrite(STBY, LOW);
}
```

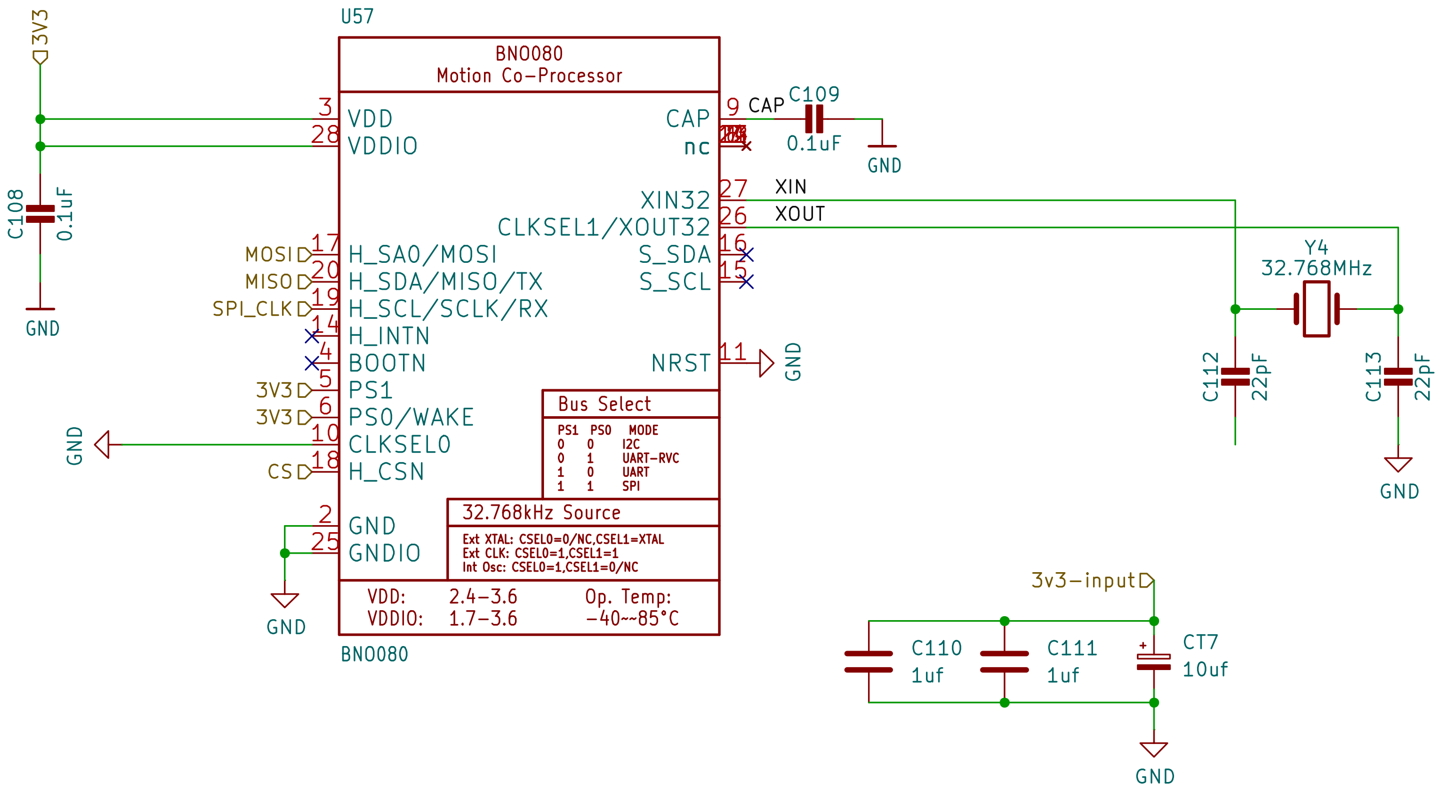

ESP_3v3
U53
MCP23S17_SO
CS SCK SI SO
CS SPI_CLK MOSI MISO
INTB INTA
RESET MCP_RST
A2 A1 A0 GND
VDD VSS
GPB0 GPB1 GPB2 GPB3 GPB4 GPB5 GPB6 GPB7
M1_EN M2_EN M3_EN M4_EN M1_DIR M2_DIR M3_DIR M4_DIR
GPA0 GPA1 GPA2 GPA3 GPA4 GPA5 GPA6 GPA7
M1_CS M2_CS M3_CS M4_CS H_BIN2 H_BIN1 H_AIN2 H_AIN1
Motor 1 power
B-B_2 Q20 IRLML2502 GND DC1 Conn_01x02_Socket
ESP_3v3
B-B_3 Q21 IRLML2502 GND DC2 Conn_01x02_Socket
USB_5V
B-B_4 Q22 IRLML2502 GND DC3 Conn_01x02_Socket
Motor 2 power
B-B_5 Q23 IRLML2502 GND DC4 Conn_01x02_Socket
ESP_3v3
B-B_6 Q24 IRLML2502 GND DC5 Conn_01x02_Socket
USB_5V
B-B_7 Q25 IRLML2502 GND DC6 Conn_01x02_Socket
ESP_3v3
MCP23S17_SO
CS SPI_CLK MOSI MISO
MCP_RST
GND 5V
SPI_CS_1 SPI_CS_2 SPI_CS_3 SPI_CS_4 SPI_CS_5 SPI_CS_6 SPI_CS_7 SPI_CS_8
H_EN I2C_RST B-B_2 B-B_3 B-B_4 B-B_5 B-B_6 B-B_7
ESP_3v3
C103 1uf
C104 1uf
C105 1uf
CT6 10uf
GND

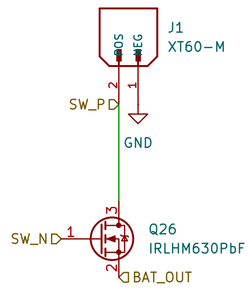

3V3
SPI_CLK
MISO
MOSI
CS8
GND
1
2
3
4
5
6
SPI1
Conn_01x06_Socket
3V3
SPI_CLK
MISO
MOSI
CS7
GND
1
2
3
4
5
6
SPI2
Conn_01x06_Socket
3V3
SPI_CLK
MISO
MOSI
CS6
GND
1
2
3
4
5
6
SPI3
Conn_01x06_Socket
3V3
SPI_CLK
MISO
MOSI
CS5
GND
1
2
3
4
5
6
SPI4
Conn_01x06_Socket
BME280
MISO
MOSI
SPI_CLK
CS1
MISO
MOSI
SPI_CLK
CS
3V3
3V3
File: BME280.kicad_sch
BNO085
MISO
MOSI
SPI_CLK
CS2
MISO
MOSI
SPI_CLK
CS
3V3
3V3
File: BNO085.kicad_sch
ICM-45686
MISO
MOSI
SPI_CLK
CS3
MISO
MOSI
SPI_CLK
CS
3V3
3V3
File: ICM-45686.kicad_sch
ADS8688
MISO
MOSI
SPI_CLK
CS4
MISO
MOSI
SPI_CLK
CS
3V3
3V3
File: ADS8688 .kicad_sch

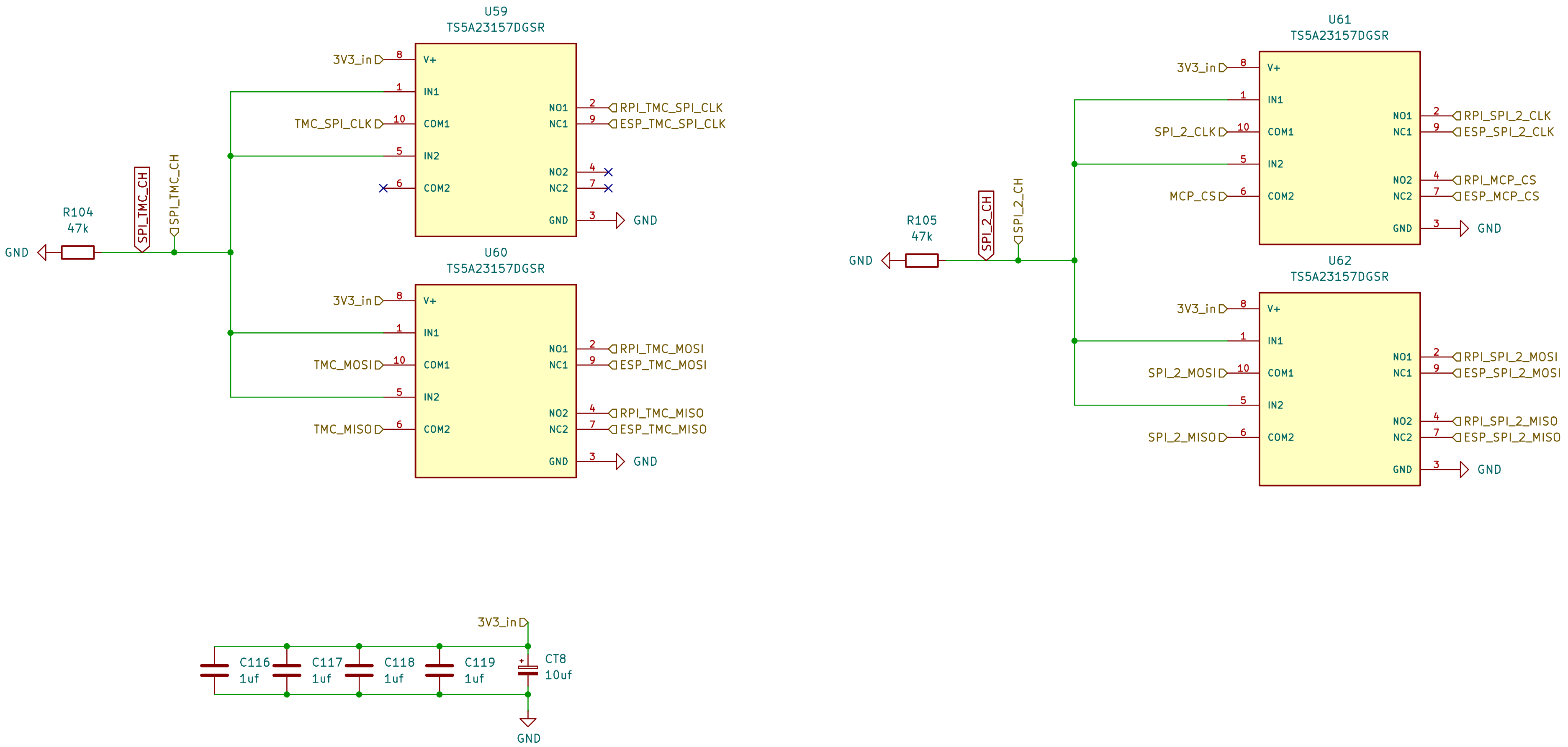


U55
ICM-45686
MOSI
SPI_CLK
CS
14
13
12
AP_SDA/AP_SDIO/AP_SDI
AP_SCL/AP_SCLK
AP_CS
MISO
1
AP_SDO/AP_AD0
2
RESV/AUX1_SDIO/AUX1_SDI/MAS_DA
3
RESV/AUX1_SCLK/MAS_CLK
4
INT1/INT
11
RESV/AUX1_SDO
10
RESV/AUX1_CS
9
INT2/FSYNC/CLKIN
8
VDD
3V3
VDDIO
GND
RESV
5
6
7
3V3
GND
C106
1uf
3V3
GND

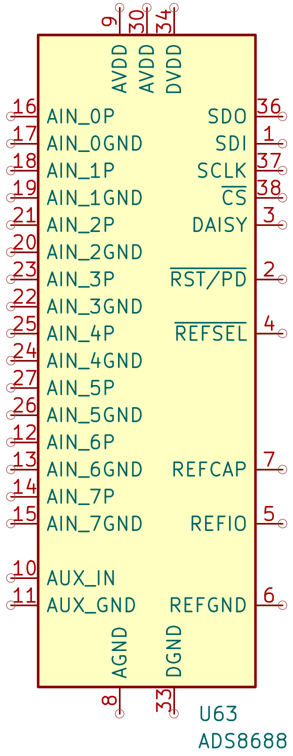

3V3
6
8
VDDIO
VDD
SDO
5
MISO
SCK
4
SPI_CLK
U56
BME280
SDI
3
MOSI
C107
1uf
GND
GND
CSB
2
CS
3V3
GND
1
7
GND
GND


U57
BNO080
Motion Co-Processor
3V3
VDD
VDDIO
CAP
nc
CAP
C109
0.1uF
GND
C108
0.1uF
GND
XIN32
CLKSEL1/XOUT32
XIN
XOUT
S_SDA
S_SCL
MOSI
MISO
SPI_CLK
H_SA0/MOSI
H_SDA/MISO/TX
H_SCL/SCLK/RX
H_INTN
BOOTN
PS1
PS0/WAKE
CLKSEL0
H_CSN
CS
NRST
Y4
32.768MHz
C112
22pF
C113
22pF
Bus Select
PS1 PS0 MODE
0 0 I2C
0 1 UART-RVC
1 0 UART
1 1 SPI
32.768kHz Source
Ext XTAL: CSEL0=0/NC,CSEL1=XTAL
Ext CLK: CSEL0=1,CSEL1=1
Int Osc: CSEL0=1,CSEL1=0/NC
GNDIO
VDD: 2.4-3.6
VDDIO: 1.7-3.6
Op. Temp: -40~85°C
3v3-input
C110
1uf
C111
1uf
CT7
10uf

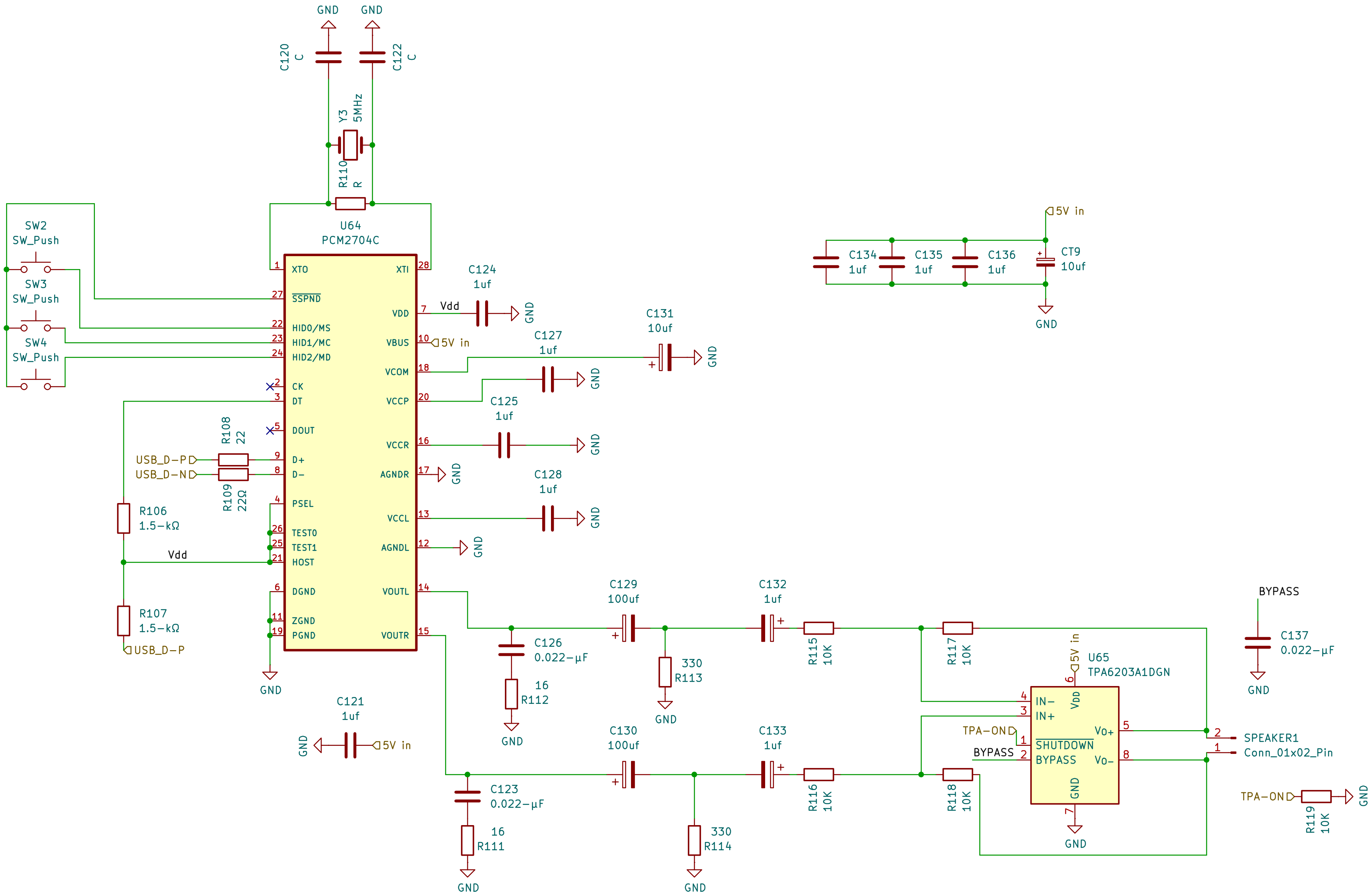

J1
XT60-M
POS
NEG
2
1
SW_P
GND
Q26
IRLHM630PbF
SW_N
1
3
2
BAT_OUT



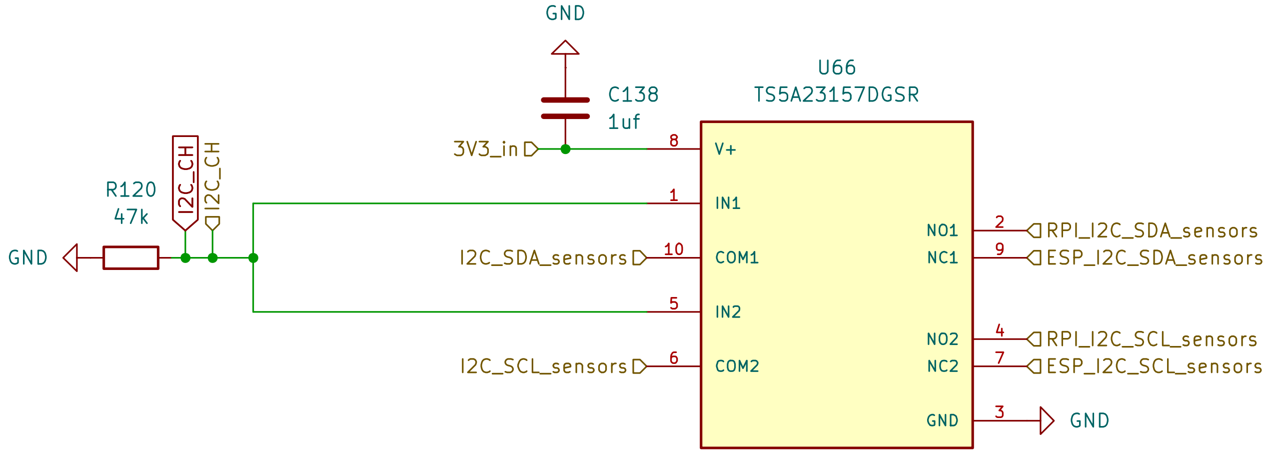
U59
TS5A23157DGSR
3V3_in
V+
IN1
TMC_SPI_CLK
COM1
IN2
COM2
NO1
NC1
NO2
NC2
GND
RPI_TMC_SPI_CLK
ESP_TMC_SPI_CLK
GND
R104
47k
GND
SPI_TMC_CH
SPI_TMC_CH
U60
TS5A23157DGSR
3V3_in
TMC_MOSI
TMC_MISO
RPI_TMC_MOSI
ESP_TMC_MOSI
RPI_TMC_MISO
ESP_TMC_MISO
GND
U61
TS5A23157DGSR
3V3_in
SPI_2_CLK
MCP_CS
RPI_SPI_2_CLK
ESP_SPI_2_CLK
RPI_MCP_CS
ESP_MCP_CS
GND
R105
47k
GND
SPI_2_CH
SPI_2_CH
U62
TS5A23157DGSR
3V3_in
SPI_2_MOSI
SPI_2_MISO
RPI_SPI_2_MOSI
ESP_SPI_2_MOSI
RPI_SPI_2_MISO
ESP_SPI_2_MISO
GND
3V3_in
C116
1uf
C117
1uf
C118
1uf
C119
1uf
CT8
10uf
GND

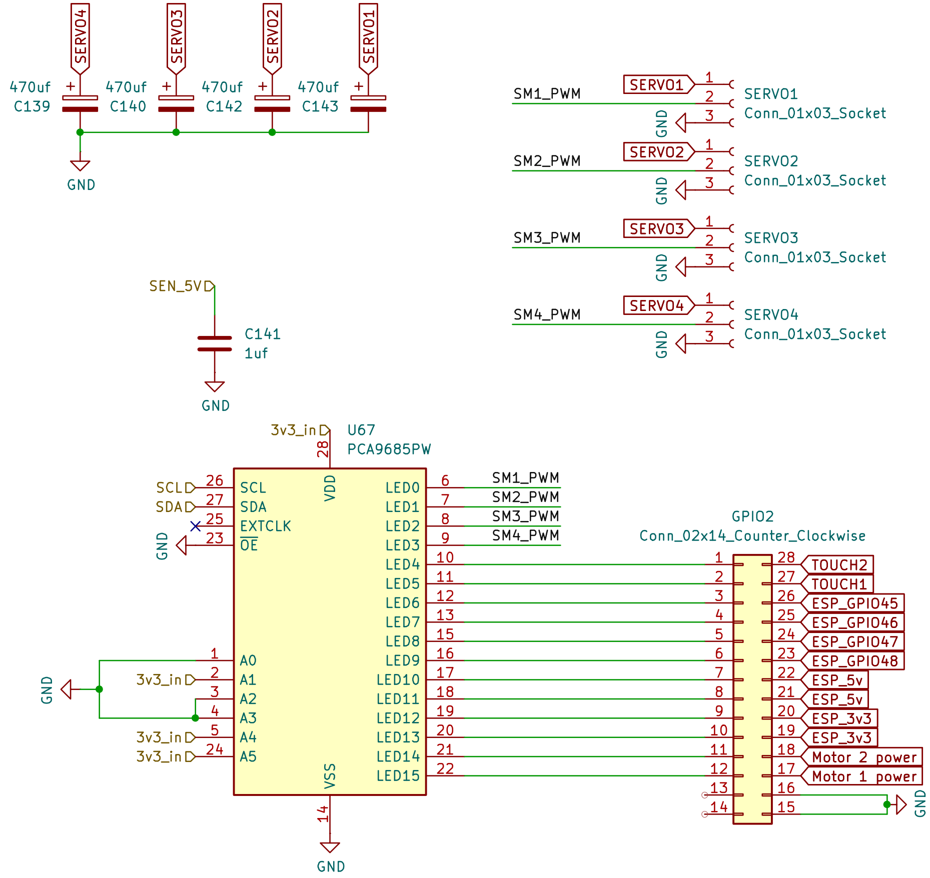


AVDD
AVDD
DVDD
AIN_0P
AIN_0GND
AIN_1P
AIN_1GND
AIN_2P
AIN_2GND
AIN_3P
AIN_3GND
AIN_4P
AIN_4GND
AIN_5P
AIN_5GND
AIN_6P
AIN_6GND
AIN_7P
AIN_7GND
AUX_IN
AUX_GND
SDO
SDI
SCLK
CS
DAISY
RST/PD
REFSEL
REFCAP
REFIO
REFGND
AGND
DGND
U63
ADS8688



GND
GND
C120
C
C122
C
Y3
5MHz
R110
R
U64
PCM2704C
SW2
SW_Push
SW3
SW_Push
SW4
SW_Push
XTO
XTI
SSPND
HID0/MS
HID1/MC
HID2/MD
CK
DT
DOUT
D+
D−
PSEL
TEST0
TEST1
HOST
DGND
ZGND
PGND
VDD
VBUS
VCOM
VCCP
VCCR
AGNDR
VCCL
AGNDL
VOUTL
VOUTR
C124
1uf
Vdd
5V in
C127
1uf
C131
10uf
C125
1uf
C128
1uf
R108
22
USB_D−P
USB_D−N
R109
22Ω
R106
1.5−kΩ
Vdd
R107
1.5−kΩ
USB_D−P
C121
1uf
5V in
C129
100uf
C132
1uf
C126
0.022−µF
16
R112
330
R113
R115
10K
R117
10K
C130
100uf
C133
1uf
C123
0.022−µF
16
R111
330
R114
R116
10K
R118
10K
5V in
U65
TPA6203A1DGN
IN−
IN+
SHUTDOWN
BYPASS
Vo+
Vo−
VDD
GND
TPA−ON
BYPASS
BYPASS
C137
0.022−µF
SPEAKER1
Conn_01x02_Pin
TPA−ON
R119
10K
5V in
C134
1uf
C135
1uf
C136
1uf
CT9
10uf
GND


GND
U66
TS5A23157DGSR
C138
1uf
3V3_in
8
V+
I2C_CH
I2C_CH
R120
47k
1
IN1
NO1
2
RPI_I2C_SDA_sensors
GND
I2C_SDA_sensors
10
COM1
NC1
9
ESP_I2C_SDA_sensors
5
IN2
NO2
4
RPI_I2C_SCL_sensors
I2C_SCL_sensors
6
COM2
NC2
7
ESP_I2C_SCL_sensors
GND
3
GND


SERVO4
SERVO3
SERVO2
SERVO1
470uf C139
470uf C140
470uf C142
470uf C143
GND
SM1_PWM
SERVO1
SERVO1 Conn_01x03_Socket
SM2_PWM
SERVO2
SERVO2 Conn_01x03_Socket
SM3_PWM
SERVO3
SERVO3 Conn_01x03_Socket
SM4_PWM
SERVO4
SERVO4 Conn_01x03_Socket
SEN_5V
C141 1uf
GND
3v3_in
U67 PCA9685PW
SCL
SDA
EXTCLK
OE
A0
A1
A2
A3
A4
A5
VDD
VSS
LED0
LED1
LED2
LED3
LED4
LED5
LED6
LED7
LED8
LED9
LED10
LED11
LED12
LED13
LED14
LED15
SM1_PWM
SM2_PWM
SM3_PWM
SM4_PWM
GPIO2
Conn_02x14_Counter_Clockwise
TOUCH2
TOUCH1
ESP_GPIO45
ESP_GPIO46
ESP_GPIO47
ESP_GPIO48
ESP_5v
ESP_5v
ESP_3v3
ESP_3v3
Motor 2 power
Motor 1 power
GND